

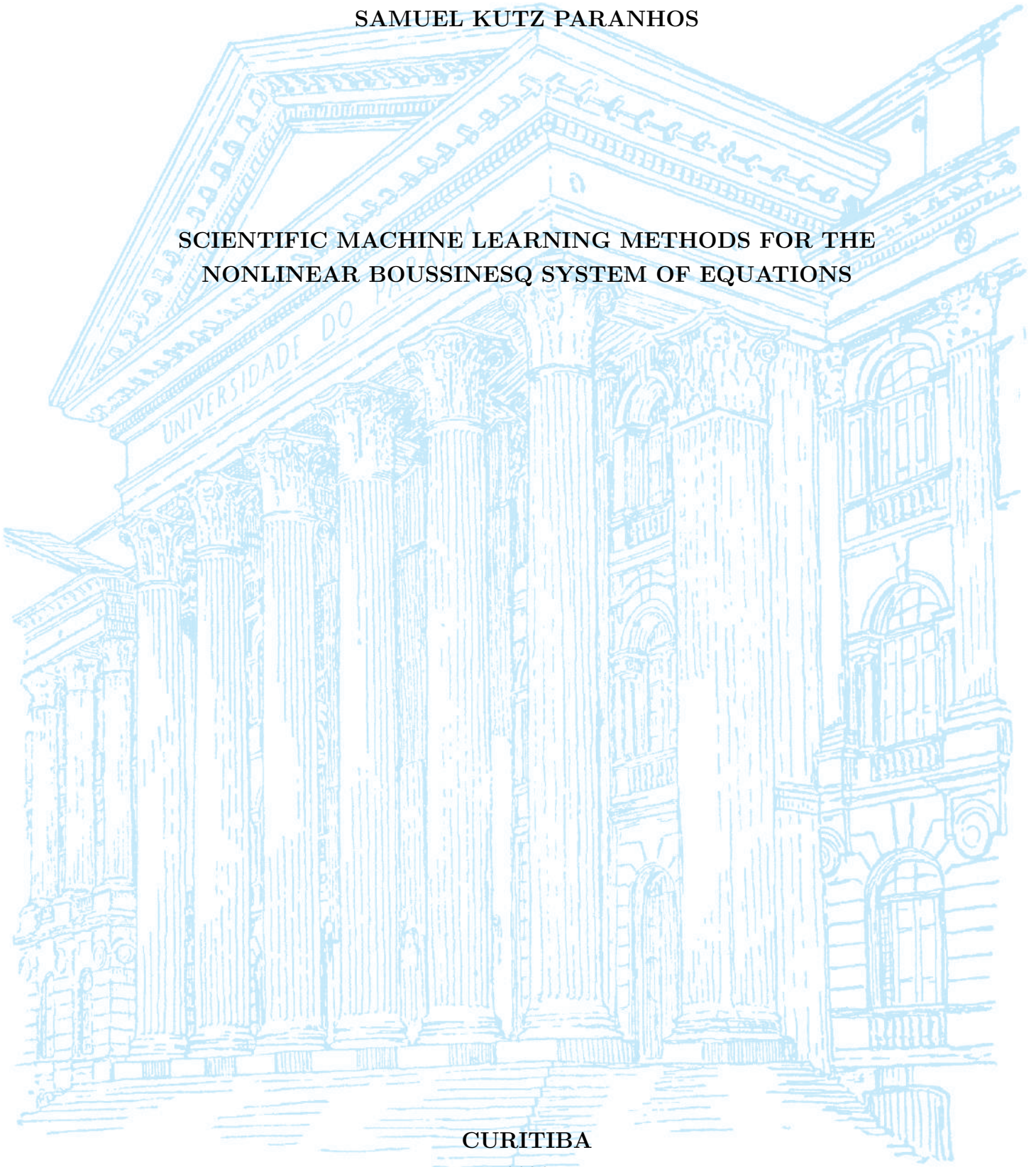
UNIVERSIDADE FEDERAL DO PARANÁ

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SCIENTIFIC MACHINE LEARNING METHODS FOR THE
NONLINEAR BOUSSINESQ SYSTEM OF EQUATIONS

CURITIBA

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TRABALHO DE CONCLUSÃO DE CURSO

**SCIENTIFIC MACHINE LEARNING METHODS FOR THE
NONLINEAR BOUSSINESQ SYSTEM OF EQUATIONS**

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RESUMO

Este trabalho implementa e avalia métodos de Aprendizado de Máquina Científico (*Scientific Machine Learning*) para o sistema de Boussinesq unidimensional, no qual o coeficiente de não linearidade α e o coeficiente de dispersão β são acoplados e variados em conjunto. Quatro arquiteturas foram implementadas do zero: um perceptron multicamadas, uma Rede Neural Informada pela Física, um Operador Neural de Fourier e um Operador Neural Informado pela Física. O cruzamento de duas famílias de arquiteturas, as redes pontuais, que representam uma única solução como um mapa contínuo das coordenadas, e os operadores, que aprendem toda uma família de parâmetros em um único treinamento, com três regimes de treinamento, apenas dados, dados e física, e apenas física, resulta em seis métodos treinados. Um método pseudoespectral com avanço temporal via Runge–Kutta de quarta ordem fornece tanto os dados supervisionados quanto a referência contra a qual todos os métodos são medidos e comparados. Um experimento mínimo à parte ajusta um único perfil de elevação de Boussinesq com redes de três larguras e mede diretamente seu Núcleo Tangente Neural (NTK) empírico, testando se a teoria atual prevê o desequilíbrio entre frequências observado ou apenas o descreve depois do fato. O viés espectral aparece em todas as arquiteturas e em todos os regimes. A presença dos dados supervisionados reduz o erro nas bandas baixa e média mais do que o resíduo da EDP, enquanto este resíduo contribui com estabilidade temporal, suprimindo o fenômeno de Gibbs no tempo final que o tratamento periódico do eixo temporal produz nos operadores. Os operadores reproduzem os números de onda baixos e médios mais de perto do que as redes pontuais e respondem a um novo parâmetro com uma única passagem direta, mas sua invariância à discretização se sustenta apenas geometricamente, sem se transferir para a acurácia: o erro é menor na malha em que o resíduo físico foi formado. As medidas do núcleo confirmam o decaimento previsto por modo enquanto o núcleo permanece fixo e localizam o viés na disparidade entre seus autovalores, e não em uma arquitetura em particular. As escalas mais finas permanecem acuradas apenas no método pseudoespectral, de modo que os métodos estudados complementam as discretizações clássicas em vez de substituí-las.

Palavras-chave: Scientific Machine Learning; Equações Diferenciais Parciais; Sistema de Boussinesq; Redes Neurais Informadas pela Física; Viés Espectral.

ABSTRACT

This work implements and evaluates Scientific Machine Learning methods for the one-dimensional Boussinesq system, in which the nonlinearity coefficient α and the dispersion coefficient β are coupled and varied together. Four architectures were written from scratch: a multilayer perceptron, a Physics-Informed Neural Network, a Fourier Neural Operator, and a Physics-Informed Neural Operator. Crossing two architecture families, the pointwise networks, which represent a single solution as a continuous map of the coordinates, and the operators, which learn a whole parameter family in one training run, with three training regimes, pure data, data and physics, and pure physics, gives six trained methods. A pseudospectral method with fourth-order Runge–Kutta time stepping supplies both the supervised data and the reference against which every method is measured and compared. A separate minimal experiment fits a single Boussinesq elevation profile with networks of three widths and measures their empirical Neural Tangent Kernel (NTK) directly, testing whether the current theory predicts the observed frequency imbalance or only describes it after the fact. Spectral bias appears in every architecture and every regime. The presence of supervised data reduces the low- and mid-band error more than the PDE residual does, while this residual contributes temporal stability, suppressing the Gibbs phenomenon at the final time that the periodic treatment of the time axis produces in the operators. The operators reproduce the low and mid wavenumbers more closely than the pointwise networks and answer a new parameter with a single forward pass, but their discretization invariance holds only geometrically and does not carry over to accuracy: the error is lowest on the grid where the physics residual was formed. The kernel measurements confirm the predicted per-mode decay while the kernel stays fixed, and locate the bias in the disparity between its eigenvalues rather than in any particular architecture. The finest scales stay accurate only in the pseudospectral method, so the methods studied here complement classical discretizations rather than replace them.

Keywords: Scientific Machine Learning; Partial Differential Equations; Boussinesq System; Physics-Informed Neural Networks; Spectral Bias.

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I chose to study applied mathematics very early in my life, around sixteen years old. I did not know a thing about it, just thought the name was cool to show around. Now, I know this is what I love. My biggest hobby. I cannot stay far from mathematics anymore and I know for a fact that for the rest of my life I will be trying to learn new mathematical content every day. It now brings food to my table, and it will continue to be a bedrock skill I developed that makes me survive in this world.

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“We are part of the universe that has developed a remarkable ability: We can hold an image of the world in our minds. We are matter contemplating itself.”

CARROLL, *The Particle at the End of the Universe*, 2012.

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LIST OF ABBREVIATIONS

SciML	Scientific Machine Learning
ML	Machine Learning
PDE	Partial Differential Equation
ODE	Ordinary Differential Equation
UDE	Universal Differential Equation
SINDy	Sparse Identification of Nonlinear Dynamics
ANN	Artificial Neural Network
MLP	Multilayer Perceptron
PINN	Physics-Informed Neural Network
FNO	Fourier Neural Operator
PINO	Physics-Informed Neural Operator
NTK	Neural Tangent Kernel
DFT	Discrete Fourier Transform
IDFT	Inverse Discrete Fourier Transform
FFT	Fast Fourier Transform
RK4	Runge–Kutta 4th-order method
GD	Gradient Descent
MSE	Mean Squared Error
AD	Automatic Differentiation
GPU	Graphics Processing Unit

LIST OF SYMBOLS

Domain and general operators

x	spatial coordinate
t	temporal coordinate
$u(x, t)$	PDE solution (generic)
$\mathcal{D}$	differential operator of the PDE
$\mathcal{L}$	linear differential operator; also a loss functional
∇	gradient (nabla) operator
Ω	spatial domain, $\Omega = [-L, L]$
T	final time of the horizon $[0, T]$
N_x	number of spatial grid points
N_t	number of time steps
$\Delta x, \Delta t$	spatial and temporal grid spacings
$\mathcal{S}$	parametric dataset of reference solutions
M	number of samples in $\mathcal{S}$

Boussinesq system and spectral quantities

$\eta(x, t)$	surface elevation field
$u(x, t)$	depth-averaged horizontal velocity field
α	nonlinearity parameter of the Boussinesq system
β	dispersion parameter of the Boussinesq system
A	initial wave amplitude
L	domain half-length, $\Omega = [-L, L]$
k	wavenumber (integer mode index)
ξ_k	wavenumber $\pi k/L$ on the domain $[-L, L]$
$\hat{f}(k)$	Fourier coefficient of f at mode k
$\hat{f}(k, t)$	time-evolving Fourier mode of f at wavenumber k
ω	primitive root of unity $e^{-2\pi i/N}$ of the DFT matrix
N_k	number of retained real-DFT modes
$B_{\text{low}}, B_{\text{mid}}, B_{\text{high}}$	low, mid and high wavenumber bands

Error metrics

$\mathbb{E}(t_n)$	time-resolved relative L^2 error at time level t_n
$\mathbb{E}_{\text{spec}}(k, t_n)$	absolute spectral error at mode k and time t_n
$\mathbb{E}_{\text{band}}(B)$	band-averaged absolute spectral error over band B

Neural networks and Neural Tangent Kernel

θ	trainable parameter vector of a neural network
u_θ	neural network approximation of the PDE solution
$h^{(l)}$	hidden-layer activation vector at layer l
σ	activation function
$W^{(l)}, b^{(l)}$	weight matrix and bias vector at layer l
N_{MLP}	number of neurons in a hidden layer
L_{MLP}	number of layers in the neural network
N_θ	number of trainable parameters
μ	learning rate / gradient-descent step size
J, J_0	Jacobian of the network output, and its value at initialization
$K(\theta), K_0$	empirical Neural Tangent Kernel, $K = JJ^\top$, and its value at initialization
V, Λ	eigenvectors and eigenvalues of K_0 , $K_0 = V\Lambda V^\top$
λ_k	NTK eigenvalue associated with mode k
λ_{max}	largest eigenvalue of K_0
v_k	k -th eigenvector of K_0
$e, e^{(n)}$	training-error vector on the grid, and its value at iteration n
c_k	error coefficient along the eigendirection v_k
τ_k	decay time constant of mode k
n_k	iterations for mode k to reach the precision ε
ε	precision threshold on the modal error
$\hat{e}(k, t)$	Fourier coefficient of the training error at mode k

Neural operators

$G^\dagger, G_\theta$	target solution operator and its neural approximation
$\mathcal{K}_\ell$	integral operator of the ℓ -th FNO spectral layer
κ_ℓ	kernel function of the FNO integral operator
P	lifting operator of the FNO
Q	projection operator of the FNO
d_c	channel (width) dimension of the FNO
$R_{\ell,k}$	FNO spectral weight (discretized kernel) at layer ℓ , mode k
$k_{\max}$	FNO mode-truncation threshold

1 INTRODUCTION

Since the start of scientific thinking, mathematics has been one of the key languages in which we can understand the world, and Partial Differential Equations (PDEs) are one of the main dialects of the mathematical language we use to model and to simplify nature’s complexity.

When modelling these equations, we cannot always afford them to have simple closed-form solutions (that is, when they have a solution at all). This exposes the need to obtain approximate solutions coming from numerical methods for these PDEs, also giving the option to circumvent analytical complexity and allowing the extraction of predictions, comparisons with data, the building of simulations, and so on. These classical numerical methods, widely studied for all types of differential equations, usually rely on discretizing the PDE’s domain into grids and performing temporal steps.

With the recent ascension of Machine Learning (ML) methods and a whole ecosystem of tools built around it, a natural path is to bridge these two areas and let ML augment the numerical solution of PDEs, taking advantage of the many software frameworks developed. This is one facet of Scientific Machine Learning (SciML), the broader effort to fuse physics-based, mechanistic models with data-driven methods.

Within SciML, this work focuses on the numerical solution of the Boussinesq System. Given the vast literature around the application of SciML on PDEs, we restrict our literature review to works on hydrodynamics equations.

1.1 Literature Overview on Scientific Machine Learning for Hydrodynamics PDEs

The neural network building blocks underlying all of these approaches trace a line beginning with the formal neuron of McCulloch et al. (1943) and Rosenblatt’s perceptron (ROSENBLATT, 1958), the first trainable single-layer architecture. The key algorithmic advance that unlocked multi-layer training was the backpropagation algorithm (RUMELHART et al., 1986), making gradient-based optimization practical for deep architectures. Shortly after, the universal approximation theorem (CYBENKO, 1989; HORNIK et al., 1989) provided the theoretical guarantee for a multilayer perceptron (MLP) with a single

hidden layer of sufficient width to be able to approximate any continuous function on a compact domain to arbitrary precision. This combination of practical trainability and universal approximation established the MLP as the backbone of modern deep learning. The term neural network itself reflects the biological metaphor at the heart of these architectures, in which McCulloch et al. (1943) drew an explicit analogy to the firing of biological neurons, and this vocabulary propagated through the field, causing MLPs to be routinely and interchangeably referred to as artificial neural networks (ANNs) or feed-forward neural networks. The idea of “deep” in deep learning, popularized following the breakthrough results of LeCun et al. (2015), simply denotes an MLP with many hidden layers, a regime in which representational capacity grows substantially and where the backpropagation-based training paradigm proves to be exceptionally efficient.

These ML foundations become the basis of a scientific methodology once the methods are coupled to the physical laws that govern a problem of interest. This is the premise of Scientific Machine Learning, a term consolidated by Baker et al. (2019) to name the discipline at the intersection of scientific computing and machine learning, and surveyed in breadth by Karniadakis et al. (2021). Its unifying goal is to combine the interpretability, generalization, and inductive biases of physics-based models with the flexibility and data-adaptivity of ML, rather than treating the network as a pure black box. The methods span several families. Some recover governing equations directly from data, as in the Sparse Identification of Nonlinear Dynamics (SINDy) (BRUNTON et al., 2016). Others embed differential-equation structure into the network itself, as in Neural Ordinary Differential Equations (neural ODEs) (CHEN, R. T. Q. et al., 2018) and Universal Differential Equations (UDEs), in which a trainable network stands in for the unknown terms of an otherwise mechanistic model (RACKAUCKAS et al., 2020). These tools have been carried across a wide range of domains, from global weather and climate forecasting (BONEV et al., 2023) to molecular dynamics, materials design, and turbulence modelling (KARNIADAKIS et al., 2021). Among all these directions, the one that concerns us here is the use of SciML to obtain numerical solutions of hydrodynamics equations, and in particular of the Boussinesq system presented in Section 1.2.

SciML for PDEs was reshaped by one idea in particular, the Physics-Informed Neural Networks (PINNs) (RAISSI et al., 2019), that is, the embedding of the governing PDE residual directly into the training loss of a Neural Network, working as a regularization term for not only fitting data, but also respecting the underlying physics. In hydrodynamics, a growing body of work has applied PINNs to water flow (DE PINA et al., 2021, 2023), shallow-water dynamics on the sphere (BIHLO et al., 2022), and inverse problems in strongly nonlinear wave systems (JAGTAP et al., 2022). The consistent result that we can address from reviews (CUOMO et al., 2023) is that PINNs are most effective when

observational data are sparse and the task is parameter identification, namely inverse problems, which are usually not well-posed. But, for a reliable precise forward simulation at scale, they do not yet rival classical solvers. In our vision, these approaches serve best to complement them.

As a natural extension from neural networks, there is a recent development in learning maps between infinite-dimensional function spaces, in which the theoretical foundation was popularized by T. Chen et al. (1995), who extended the universal approximation theorem to nonlinear operators, in other words, a shallow network can approximate any continuous operator between Banach spaces to arbitrary precision. Bringing this idea into practical terms, Lu et al. (2021) introduced DeepONet, the first deep architecture expressly designed for learning operators coming from differential equations, demonstrating it across a range of ODEs and PDEs. Building on this line of work, operator learning puts aside the single-instance limitation of PINNs by training an operator to approximate the full mapping from parameters and initial data to the solution, so that at inference time a new solution costs only a forward pass for the network. Parallel to DeepONet, Fourier Neural Operator (FNO) (LI et al., 2021) is one of the most widely adopted instances of this idea: it parametrizes an integral kernel in Fourier space, achieving speed-ups over other neural operators, and it is discretization-independent, the so-called zero-shot superresolution. Subsequent works explored alternative bases like wavelets (GUPTA et al., 2021), but FNO was validated on many types of equations such as dispersive and soliton wave equations (PEI et al., 2022; ZHONG et al., 2023), and was also scaled to geophysical problems spanning regional ocean modeling, shallow-water emulation, and global weather forecasting (BONEV et al., 2023; LKHAGVASUREN et al., 2024; UCHIDA et al., 2025; PATEL et al., 2025; YU et al., 2025).

Following the idea for PINNs, the Physics-Informed Neural Operator (PINO) (LI et al., 2023a) combines operator learning with PDE-residual constraints, cutting the labelled-data requirement while preserving physical consistency. Applications to shallow-water systems (ROSOFISKY et al., 2023) confirmed that this hybrid regime inherits the generalization of FNO without its data hunger. Broad surveys of the field (TODO, 2023) identify these physics-data approaches working together as a promising SciML direction, while naming turbulence modeling, scalability, and noisy-data robustness as open challenges.

A known limitation that cuts across all SciML architectures is the so-called spectral bias, the tendency of gradient-based optimizers to learn the low-frequency components of a solution faster than the high-frequency ones (RAHAMAN et al., 2019; XU et al., 2020). The theoretical account of this behavior came with the Neural Tangent Kernel (NTK) (JACOT et al., 2018), which showed that a sufficiently wide network trains, to

leading order, as a linear model governed by a kernel that stays essentially fixed during optimization, the so-called lazy-training regime. In this picture each mode of the solution is learned at a rate set by its associated NTK eigenvalue, so the wide disparity between the largest and smallest eigenvalues translates directly into the frequency imbalance seen during training (ARORA et al., 2019; CAO et al., 2021; BORDELON et al., 2020). Wang, Yu, et al. (2022) carried this analysis to PINNs, showing that the eigenvalue disparity between the PDE-residual and initial-condition loss components is what drives their spectral bias. This pathology is especially consequential for dispersive wave problems, where the solution energy is broadly distributed across wavenumbers. A recent systematic study (KHODAKARAMI et al., 2026) showed that second-order optimizers combined with spectrum-aware loss formulations substantially mitigate it.

As far as this review reveals, SciML methods have been applied to a broad class of shallow-water and dispersive wave equations. To the best of our knowledge, however, no study has systematically evaluated FNO, PINO, or PINNs on the parametric 1D Boussinesq system, where both the nonlinearity coefficient α and the dispersion coefficient β vary simultaneously. This gap motivates our effort to implement and evaluate these SciML methods on that system, documenting the details a reader would need to apply them to a dispersive wave problem of their own.

1.2 The Boussinesq System of Equations

For this work, we choose to implement SciML methods for a specific PDE originating from J. Boussinesq’s 1872 effort to give a theoretical account of the solitary wave of translation observed by J. Scott Russell in a canal in 1834 (BOUSSINESQ, 1872). Russell had noticed that a hump of water, set in motion by a stopping barge, could travel for miles without changing shape, a phenomenon the linear wave theory of the time could not explain. Boussinesq showed that the stability of such a wave is the result of a precise balance between nonlinear steepening and frequency dispersion, two effects that are separately destabilizing but mutually compensating in the shallow-water, long-wave regime. The mathematical structure of this balance, and of dispersive wave theory in general, are treated in depth by Whitham (1974) and Debnath (2012).

A modern two-field form of the system, for the free-surface elevation $\eta(x, t)$ and the depth-averaged horizontal velocity $u(x, t)$, was systematically derived and classified by Bona et al. (2002), who identified a four-parameter family of equivalent Boussinesq systems obtainable from the incompressible, irrotational Euler equations by asymptotic simplifications. Earlier, Peregrine (1967) had already written down the standard form of these equations for water of variable depth, establishing the notation and scaling that remain in widespread use. What makes the Boussinesq system particularly suitable as

a parametric testbed for this work is that changing the nonlinearity and dispersion coefficients continuously deforms the solution from the near-linear, wave-packet regime to the strongly nonlinear soliton regime, giving rise to a more interesting range of behaviors within a single mathematical framework.

The specific one-dimensional form of the system used in this work, with its parameterization of the nonlinearity coefficient α and the dispersion coefficient β , was adopted from Martins (2023) and Martins et al. (2024), which is:

$$\begin{cases} \eta_t + u_x + \alpha(\eta u)_x = 0, \\ u_t - \frac{\beta}{3}u_{xxt} + \eta_x + \alpha uu_x = 0. \end{cases} \quad (1.1)$$

For this work, the parameters α and β in (1.1) are held equal, defining a one-parameter family of problems whose difficulty grows with the parameter. Namely, for small $\alpha = \beta$ the dynamics are nearly linear and the solution is well-captured by low-frequency modes, while for large $\alpha = \beta$ the strong coupling between nonlinearity and dispersion produces richer dynamics that are fundamentally harder to learn. This controlled variation is precisely what makes the setting useful for comparing SciML architectures, with the goal of observing how each method learns the target solution’s frequency.

Classical numerical methods for the Boussinesq system rely on pseudospectral and finite-difference discretizations that exploit the periodic or nearly periodic spatial structure of the waves. Pseudospectral (TREFETHEN, 2000) schemes achieve spectral accuracy on the spatial derivatives and are therefore well suited to the dispersive character of the equations (STEINMOELLER et al., 2012); this is the approach used to generate the reference dataset in this work. On the application side, operational coastal-engineering codes such as FUNWAVE (TODO, 2010) implement Boussinesq-type solvers for harbor and nearshore wave propagation, demonstrating the practical relevance of the model beyond the academic setting.

SciML methods have recently been applied only to the Boussinesq equation, not to the specific Boussinesq system. Liu et al. (2023) applied parameter-integrated PINNs with phase-domain decomposition to reconstruct two-dimensional Boussinesq dynamics including phase transitions and rogue-wave formation, identifying spectral bias as a key obstacle at high nonlinearity. Wang et al. (2024) trained a physics-informed network to infer velocity and pressure fields from surface-elevation data alone on a Boussinesq model, demonstrating the inverse-problem capabilities of the approach. Complementary studies on related nonlinear wave equations (ZHANG et al., 2024; KUMAR et al., 2025) have confirmed that purely data-driven and purely physics-driven networks each display characteristic failure modes at strong nonlinearity. Taken together, these results suggest

that a systematic, parametric evaluation of FNO/PINO/PINN on the 1D Boussinesq system, which is the focus of this work, can be fruitful to understand better the interactions between the methods’ ease of training and nonlinearities and dispersiveness.

1.3 Research Goals

This work implements and evaluates four SciML architectures on the parametric 1D Boussinesq system, arranged into two families of three trained methods each. The neural networks family represents a single parameter: a plain multilayer perceptron (MLP) trained on data alone, and a Physics-Informed Neural Network (PINN) trained both with and without supervised data. The operator family learns the whole parameter range in a single training and answers a new parameter with one forward pass: a Fourier Neural Operator (FNO) trained on data alone, and a Physics-Informed Neural Operator (PINO) trained both with and without supervised data. Crossing the two families with the three training regimes, pure data, data and physics, and pure physics, gives the six methods compared throughout. A separate strand runs beside them: a minimal network fitting a fixed time Boussinesq elevation profile, whose Neural Tangent Kernel (NTK) is measured directly to isolate the spectral-bias mechanism. All results are measured against a pseudospectral reference solver, which sets a high-accuracy baseline. The study is guided by the following questions.

- Accuracy relative to the pseudospectral baseline: how closely can each family reproduce the reference solutions as nonlinearity and dispersion grow? Where does each method start to fail, and how does its error scale with the difficulty of the regime? For the operators, does the discretization invariance their spectral layers promise survive a query off the training grid?
- Data-driven, physics-informed, and hybrid training: how does the presence or absence of supervised reference data affect the accuracy and training behavior of each method? Does adding a physics residual term improve generalization, or does it introduce new difficulties?
- Spectral bias across training regimes: how does spectral bias manifest in each architecture when the error spectrum is tracked band by band during training, and to what extent does the inclusion of data or physics constraints alter the frequency at which errors concentrate?
- The mechanism behind spectral bias: does Neural Tangent Kernel (NTK) theory genuinely predict the frequency imbalance seen on the Boussinesq system, or only describe it after the fact? We put the theory to two quantitative tests on a minimal

network fitting a single Boussinesq elevation profile at a fixed time, across a range of network widths. The first targets the initial-kernel (lazy-training) assumption the analysis depends on: we measure how far the network parameters and the empirical kernel drift over training, and whether the kernel’s eigenvalue spectrum is essentially unchanged from initialization to final iteration. The second targets the predicted driver of the bias: modes carrying the high-frequency detail are learned far more slowly than low-frequency ones. Testing both, where they hold and where they break, ties the frequency-band errors seen during training to a concrete mechanism instead of a loose analogy.

Together, these questions locate where each family and each training regime holds up on dispersive, nonlinear wave problems and where it fails, and what that implies for choosing among them.

1.4 Work Structure

This work is written with the goal of introducing SciML to senior undergraduates in mathematics, physics, or engineering who aim to apply these methods in their future research, presenting its pros and cons. We assume the reader has good knowledge of scientific computing and linear algebra. The remaining chapters are organized as follows.

- Chapter 2 develops the mathematical background. It covers the Boussinesq system and its analytical properties, the discrete Fourier transform and the pseudospectral solver built on it, and then the theoretical foundations of the two families, the MLP and the PINN on one side and the FNO and the PINO on the other. It closes with the Neural Tangent Kernel and a derivation of the spectral bias mechanism that the later chapters put to the test.
- Chapter 3 describes the experimental methodology. It details the computational setup, the pseudospectral dataset that serves as both training target and ground truth, the error metrics that score every method on one scale, the six trained methods on the two-family by three-regime grid, and the minimal experiment that isolates the spectral-bias mechanism.
- Chapter 4 presents the numerical results. It accounts for what each method costs to train, then takes them along one axis at a time: the nonlinearity tests, the multi-resolution tests, and the spectrum compared mode by mode and closes with the NTK measurements and the frequency-band tracking that tie the observed errors to spectral bias for all methods.

- Chapter 5 assesses each family against the questions posed above, states what the study cannot claim, and outlines directions for future work.

2 MATHEMATICAL BACKGROUND

This chapter introduces the mathematical foundations of the methods studied in this work. Section 2.1 presents the Boussinesq system and the limiting wave-equation regime. Section 2.2 develops the pseudospectral numerical method used to generate reference solutions. Sections 2.3 and 2.4 introduce, respectively, the neural network and neural operator architectures studied in this work.

2.1 The Boussinesq System

The Boussinesq system is a weakly nonlinear, weakly dispersive model governing the coupled evolution of two scalar fields in shallow water: the free surface elevation $\eta(x, t)$ and the depth-averaged horizontal velocity $u(x, t)$, both depending on position x and time t . In the one-dimensional, spatially periodic setting studied in this work, the system reads

$$\eta_t + u_x + \alpha (\eta u)_x = 0, \quad (2.1a)$$

$$u_t - \frac{\beta}{3} u_{xxt} + \eta_x + \alpha u u_x = 0, \quad (2.1b)$$

on the spatial domain $x \in [-L, L]$, where $\alpha \geq 0$ is the nonlinearity parameter, $\beta > 0$ is the dispersion parameter, and $L > 0$ is the domain half-length. The two coefficients weigh the competing effects the model balances: α scales the nonlinear terms $(\eta u)_x$ and $u u_x$, which steepen the wave, while β scales the dispersive term u_{xxt} , which spreads it. This nondimensional form follows equation (3.1) of Martins (2023) (see also Martins et al. (2024)).

The system (2.1) is solved throughout this work from a single solitary profile at rest,

$$\eta(x, 0) = A \operatorname{sech}^2(x), \quad u(x, 0) = 0, \quad (2.2)$$

where $A > 0$ is the initial amplitude and $\operatorname{sech}(z) = 2/(e^z + e^{-z})$ is the hyperbolic secant. This places one crest at the centre $x = 0$ of an otherwise flat surface, with zero initial velocity, as drawn in Figure 2.1.

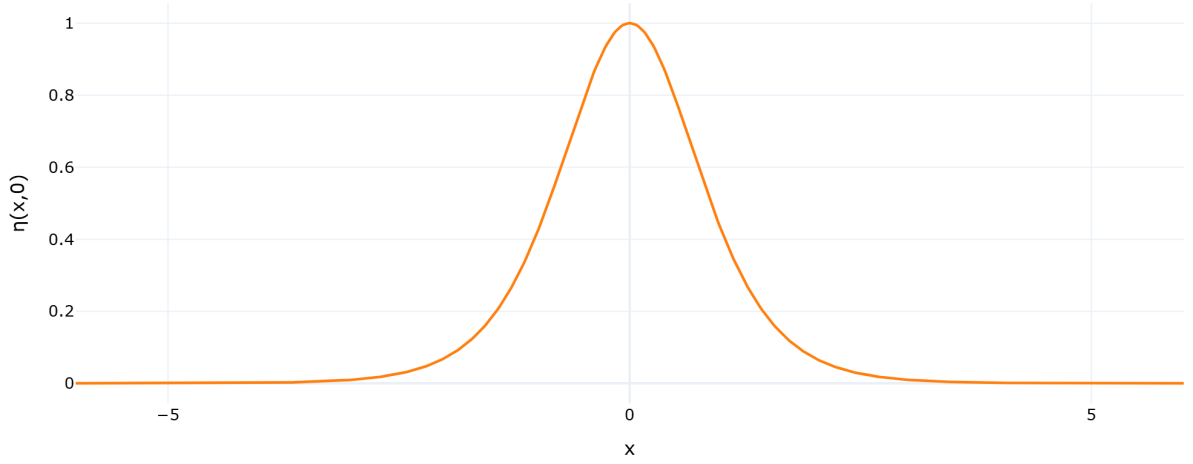


Figure 2.1 – Initial condition $\eta(x, 0) = A \operatorname{sech}^2(x)$ for $A = 1$.

Source: The author (2026).

2.1.1 Recovering the Wave Equation

When both coefficients vanish, $\alpha = \beta = 0$, the nonlinear and the dispersive terms disappear together and the Boussinesq system reduces to:

$$\eta_t + u_x = 0, \quad (2.3)$$

$$u_t + \eta_x = 0. \quad (2.4)$$

The two fields decouple into a single scalar equation. Differentiating (2.3) with respect to t gives

$$\eta_{tt} + u_{xt} = 0, \quad (2.5)$$

and differentiating (2.4) with respect to x gives

$$u_{tx} + \eta_{xx} = 0. \quad (2.6)$$

Since $u_{xt} = u_{tx}$, subtracting (2.6) from (2.5) cancels the mixed derivative and leaves

$$\eta_{tt} - \eta_{xx} = 0,$$

that is, the wave equation for η ,

$$\eta_{tt} = \eta_{xx}. \quad (2.7)$$

In an analogous manner, differentiating (2.4) with respect to t and (2.3) with respect to x , the same cancellation leaves $u_{tt} - u_{xx} = 0$, so u obeys the same equation.

D'Alembert's formula (DEBNATH, 2012) provides the general solution to (2.7).

Given initial data $\eta(x, 0) = f(x)$ and $\eta_t(x, 0) = g(x)$, the solution is

$$\eta(x, t) = \frac{f(x+t) + f(x-t)}{2} + \frac{1}{2} \int_{x-t}^{x+t} g(s) ds. \quad (2.8)$$

We now apply this to the initial condition (2.2). From (2.3), $\eta_t(x, 0) = -u_x(x, 0) = 0$, so $f(x) = A \operatorname{sech}^2(x)$ and $g \equiv 0$. The integral in (2.8) vanishes and the surface elevation evolves as

$$\eta(x, t) = \frac{A}{2} [\operatorname{sech}^2(x+t) + \operatorname{sech}^2(x-t)]. \quad (2.9)$$

The initial bump of amplitude A therefore splits into two counter-propagating pulses of amplitude $A/2$, travelling at speed ± 1 without changing shape.

For u , from (2.4) we have $u_t(x, 0) = -\eta_x(x, 0)$, so $f \equiv 0$ and $g(x) = -\eta_x(x, 0)$. The f term vanishes in (2.8), leaving

$$u(x, t) = -\frac{1}{2} \int_{x-t}^{x+t} \eta_x(s, 0) ds = -\frac{1}{2} [\eta(x+t, 0) - \eta(x-t, 0)],$$

and substituting $\eta(x, 0) = A \operatorname{sech}^2(x)$,

$$u(x, t) = \frac{A}{2} [\operatorname{sech}^2(x-t) - \operatorname{sech}^2(x+t)]. \quad (2.10)$$

Any deviation from this wave-equation baseline in the full system (2.1) is attributable either to dispersion ($\beta > 0$) or to nonlinearity ($\alpha > 0$).

2.2 Pseudospectral Method

The pseudospectral method is applied here to the Boussinesq system (2.1) on the periodic spatial domain $\Omega = [-L, L]$. It transforms the two fields in space alone and leaves time as a separate evolution direction, which is the treatment an evolution equation asks for: taking the DFT along x turns each spatial derivative into an algebraic multiplication, so the PDE collapses into a system of ordinary differential equations in time, one for every spatial frequency.

2.2.1 Spatial Discretization and the Discrete Fourier Transform

The spatial domain $\Omega = [-L, L]$ of total length $2L$ is discretized into N_x equispaced points:

$$x_j = -L + j \Delta x, \quad \Delta x = \frac{2L}{N_x}, \quad j = 0, \dots, N_x - 1.$$

The Discrete Fourier Transform (DFT) of a function f sampled at these points is (TREFETHEN, 2000):

$$\hat{f}(k) = \sum_{j=0}^{N_x-1} f(x_j) e^{-2\pi i k j / N_x}, \quad k = -\frac{N_x}{2}, \dots, \frac{N_x}{2} - 1, \quad (2.11)$$

with its inverse, the Inverse Discrete Fourier Transform (IDFT), recovering the physical values:

$$f(x_j) = \frac{1}{N_x} \sum_k \hat{f}(k) e^{2\pi i k j / N_x}. \quad (2.12)$$

The fundamental identity of the DFT with respect to differentiation is that, for a periodic function, the spatial derivative transforms into a pointwise multiplication (TREFETHEN, 2000; STEINMOELLER et al., 2012):

$$\widehat{(f_x)}(k) = i \frac{\pi k}{L} \hat{f}(k), \quad \widehat{(f_{xx})}(k) = -\frac{\pi^2 k^2}{L^2} \hat{f}(k).$$

This is in contrast to finite-difference methods, where the derivative carries a truncation error that shrinks with the grid spacing.

We define the wavenumber of mode k on $[-L, L]$ as:

$$\xi_k := \frac{\pi k}{L}. \quad (2.13)$$

With this definition, the derivative identities read:

$$\widehat{(f_x)}(k) = i \xi_k \hat{f}(k), \quad \widehat{(f_{xx})}(k) = -\xi_k^2 \hat{f}(k). \quad (2.14)$$

2.2.2 Spectral Formulation of the Boussinesq System

To bring the Boussinesq system (2.1) into the Fourier domain, following the spectral formulation of Martins (2023), we apply the DFT to each equation and use the derivative identities (2.14). Equation (2.1a), treating $\widehat{(\eta u)}(k, t)$ as a given nonlinear term, becomes:

$$\hat{\eta}_t(k, t) = -i \xi_k \hat{u}(k, t) - \alpha i \xi_k \widehat{(\eta u)}(k, t). \quad (2.15)$$

Equation (2.1b) requires more care because of the u_{xxt} term. Since spatial derivatives commute with the DFT:

$$\widehat{(u_{xxt})}(k, t) = -\xi_k^2 \hat{u}_t(k, t).$$

Substituting into the DFT of (2.1b), with its nonlinear term written in the equivalent form $uu_x = \frac{1}{2}(u^2)_x$:

$$\hat{u}_t(k, t) + \frac{\beta}{3} \xi_k^2 \hat{u}_t(k, t) + i \xi_k \hat{\eta}(k, t) + \alpha i \xi_k \widehat{\left(\frac{1}{2}u^2\right)}(k, t) = 0.$$

Collecting the $\hat{u}_t(k, t)$ terms on the left and solving algebraically:

$$\hat{u}_t(k, t) = \frac{-i \xi_k \hat{\eta}(k, t) - \alpha i \xi_k \widehat{\left(\frac{1}{2}u^2\right)}(k, t)}{1 + \frac{\beta \xi_k^2}{3}}. \quad (2.16)$$

The division in (2.16) is where the transform pays off. Isolating $\hat{u}_t$ means inverting the operator $1 - \frac{\beta}{3} \frac{\partial^2}{\partial x^2}$, which on the grid couples neighbouring nodes; under the DFT it is diagonal, acting on mode k as multiplication by $1 + \frac{\beta \xi_k^2}{3}$, so the inversion reduces to one division per mode.

Equations (2.15) and (2.16) can be written jointly as a first-order ODE system in time for each fixed wavenumber $k \in \mathbb{Z}$:

$$\begin{cases} \hat{\eta}_t(k, t) = -i \xi_k \hat{u}(k, t) - \alpha i \xi_k \widehat{(\eta u)}(k, t), \\ \hat{u}_t(k, t) = \frac{-i \xi_k \hat{\eta}(k, t) - \alpha i \xi_k \widehat{\left(\frac{1}{2}u^2\right)}(k, t)}{1 + \frac{\beta \xi_k^2}{3}}. \end{cases} \quad (2.17)$$

The PDE system (2.1), two equations in (x, t) , is thus equivalent to an infinite family of two-dimensional ODE systems (2.17), one for each $k \in \mathbb{Z}$, coupled through the nonlinear terms.

The nonlinear terms $\widehat{(\eta u)}(k, t)$ and $\widehat{\left(\frac{1}{2}u^2\right)}(k, t)$ cannot be computed directly in Fourier space without a convolution sum, which costs $O(N_x^2)$. The pseudospectral approach avoids this by computing products in physical space. For $\widehat{(\eta u)}(k, t)$, the procedure is:

1. recover $\eta(x_j)$ and $u(x_j)$ by applying the inverse DFT (2.12);
2. compute the product $(\eta u)_j = \eta(x_j) \cdot u(x_j)$ pointwise on the grid;
3. apply the DFT (2.11) to obtain $\widehat{(\eta u)}(k, t)$.

Similarly, $\widehat{\left(\frac{1}{2}u^2\right)}(k, t)$ is obtained by the same three-step procedure with $u^2(x_j) = u(x_j)^2$, where $u(x_j) = \text{IDFT}(\hat{u})(x_j)$; and $(\eta u)(x_j) = \text{IDFT}(\hat{\eta})(x_j) \cdot \text{IDFT}(\hat{u})(x_j)$. Since each DFT/IDFT costs $O(N_x \log N_x)$ via the Fast Fourier Transform (FFT), this pseudospectral strategy assembles each nonlinear term in $O(N_x \log N_x)$ instead of $O(N_x^2)$.

The nonlinear term of (2.1b) fits this procedure only in the form it was given in (2.16). Since $uu_x = \frac{1}{2}(u^2)_x$, the two forms are equivalent, but only the second is a product of a field with itself: recovering u on the grid, squaring it pointwise and transforming back costs one inverse and one forward transform, whereas uu_x would require recovering u and u_x separately before multiplying them. Both forms appear in this work. The solver uses $\frac{1}{2}(u^2)_x$ for exactly this reason, while the physics-informed network of Section 2.3.5 differentiates uu_x directly by automatic differentiation, where the products are never assembled on a grid and the distinction carries no cost.

2.2.3 Time Evolution via Runge-Kutta 4

Collecting the two spectral fields into a single state vector at each wavenumber k ,

$$\hat{Y}(k, t) = (\hat{\eta}(k, t), \hat{u}(k, t)),$$

the system (2.17) takes the standard form of an initial value problem for a system of ordinary differential equations,

$$\hat{Y}_t(k, t) = F(\hat{Y}(k, t)), \quad (2.18)$$

where the pseudospectral right-hand side operator F is the map

$$F(\hat{Y}(k, t)) = F(\hat{\eta}, \hat{u})(k, t) = \begin{pmatrix} -i \xi_k \hat{u}(k, t) - \alpha i \xi_k \widehat{(\eta u)}(k, t) \\ \frac{-i \xi_k \hat{\eta}(k, t) - \alpha i \xi_k \widehat{(\frac{1}{2} u^2)}(k, t)}{1 + \frac{\beta \xi_k^2}{3}} \end{pmatrix},$$

whose two components are $\hat{\eta}_t$ and $\hat{u}_t$ as given by equations (2.15)–(2.16), with the nonlinear terms assembled by the pseudospectral procedure above.

In the form (2.18) the problem is what any explicit one-step integrator expects, and time evolution uses the classical fourth-order Runge-Kutta (RK4) scheme (SÜLI et al., 2003; BURDEN et al., 2011). Discretizing (2.18) in time, write $\hat{Y}_n = (\hat{\eta}^n, \hat{u}^n)$ for the spectral state at time t_n and fixed wavenumber k . The four stage evaluations $\kappa_1, \kappa_2, \kappa_3, \kappa_4$

and the update to $t_{n+1} = t_n + \Delta t$ are:

$$\begin{aligned}\kappa_1 &= F(\hat{Y}_n), \\ \kappa_2 &= F\left(\hat{Y}_n + \frac{\Delta t}{2} \kappa_1\right), \\ \kappa_3 &= F\left(\hat{Y}_n + \frac{\Delta t}{2} \kappa_2\right), \\ \kappa_4 &= F\left(\hat{Y}_n + \Delta t \kappa_3\right), \\ \hat{Y}_{n+1} &= \hat{Y}_n + \frac{\Delta t}{6} (\kappa_1 + 2\kappa_2 + 2\kappa_3 + \kappa_4).\end{aligned}$$

At each stage, the intermediate state is decoded to physical space to assemble the nonlinear products, then encoded back to Fourier space to evaluate the spectral derivatives. This combination of physical-space products, Fourier-space derivatives, and time evolution by a separate marching scheme is what characterizes a pseudospectral method.

2.2.4 Training Dataset

The numerical experiments in this work vary only the PDE parameters (α, β) while keeping the initial condition fixed at (2.2). The methods are therefore trained and evaluated on the dataset

$$\mathcal{S} = \left\{ ((\alpha_i, \beta_i), (\eta_i, u_i)) \right\}_{i=1}^M, \quad (2.19)$$

where (α_i, β_i) are the nonlinearity and dispersion parameters for the i -th sample, and (η_i, u_i) is the corresponding space-time solution obtained from the pseudospectral solver of Section 2.2 with those parameters and the shared initial condition (2.2).

2.3 Neural Networks

This section introduces the neural network architectures used in this work, together with the Neural Tangent Kernel theory that describes how such networks train. We begin with the Multilayer Perceptron, the building block underlying every architecture studied here. We then develop the Neural Tangent Kernel, the tool we use to analyse the spectral bias a network produces when it is fitted to data sampled on a regular grid. Finally we turn to Physics-Informed Neural Networks, where the same kernel analysis carries over and spectral bias is a well-documented obstacle.

2.3.1 Multilayer Perceptrons

In this work, we focus on Multilayer Perceptrons (MLPs), popularized by Rumelhart et al. (1986). These networks are compositions of affine transformations parametrized

by tunable weights and biases, separated by nonlinear activations, where an optimization process fits the network to a dataset.

Given a labeled dataset $\{(x_i, u_i)\}_{i=1}^N$, the objective is to fit the function $u_\theta(x)$ by minimizing a loss function, here the Mean Squared Error (MSE). With $\|\cdot\|$ denoting the Euclidean norm, the unconstrained optimization problem is:

$$\mathcal{L}_{\text{MLP}}(\theta) = \frac{1}{N} \sum_{i=1}^N \|u_\theta(x_i) - u_i\|^2. \quad (2.20)$$

The MLP $u_\theta : \mathbb{R}^{d_{\text{in}}} \rightarrow \mathbb{R}^{d_{\text{out}}}$ is defined by the composition:

$$\begin{aligned} h^{(0)} &= x \in \mathbb{R}^{d_{\text{in}}}, \\ h^{(l)} &= \sigma(W^{(l)}h^{(l-1)} + b^{(l)}) \in \mathbb{R}^{N_{\text{MLP}}}, \quad l = 1, \dots, L_{\text{MLP}} - 1, \\ u_\theta(x) &= W^{(L_{\text{MLP}})}h^{(L_{\text{MLP}}-1)} + b^{(L_{\text{MLP}})} \in \mathbb{R}^{d_{\text{out}}}. \end{aligned}$$

where:

- $L_{\text{MLP}} \in \mathbb{N}$ is the number of layers and $N_{\text{MLP}} \in \mathbb{N}$ the number of neurons per hidden layer;
- $W^{(l)} \in \mathbb{R}^{N_{\text{MLP}} \times N_{\text{MLP}}}$ and $b^{(l)} \in \mathbb{R}^{N_{\text{MLP}}}$ are the weight matrix and bias vector at layer l , with $W^{(1)} \in \mathbb{R}^{d_{\text{in}} \times N_{\text{MLP}}}$ and $W^{(L_{\text{MLP}})} \in \mathbb{R}^{N_{\text{MLP}} \times d_{\text{out}}}$ adapted at the boundaries;
- $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a sigmoidal activation function applied elementwise, satisfying $\sigma(t) \rightarrow 1$ as $t \rightarrow +\infty$ and $\sigma(t) \rightarrow 0$ as $t \rightarrow -\infty$;
- $\theta = \{W^{(l)}, b^{(l)}\}_{l=1}^{L_{\text{MLP}}}$ denotes all trainable parameters.

If the activation function is sigmoidal, MLPs are universal approximators: they can approximate any continuous function on a compact domain to arbitrary precision (CYBENKO, 1989).

In Figure 2.2, we show a common depiction of the neural network. Here the input vector is the pair $(x, t) \in \mathbb{R}^2$ and the output is $(\eta_\theta(x, t), u_\theta(x, t)) \in \mathbb{R}^2$.

2.3.2 Gradient Flow

The analysis of the following sections is carried out on the continuous-time limit of gradient descent, the form in which the neural-tangent-kernel results this work relies on are stated (JACOT et al., 2018; WANG; YU, et al., 2022). This subsection derives that limit.

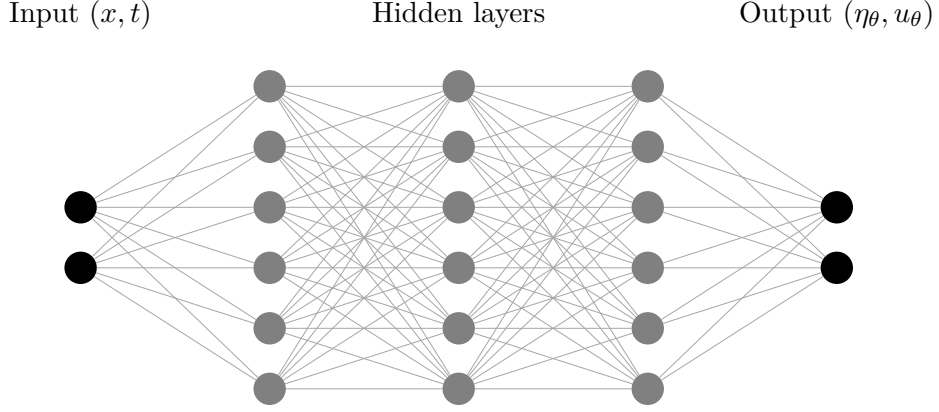


Figure 2.2 – Example of MLP architecture of input (x, t) , with $L_{\text{MLP}} = 3$ hidden layers and $N_{\text{MLP}} = 6$ neurons per layer.

Source: The author (2026).

To understand how an MLP minimizes its data-fitting loss (2.20), consider gradient descent with learning rate $\mu > 0$. Starting from an initialization $\theta^{(0)}$, each iteration updates the parameters by

$$\theta^{(k+1)} = \theta^{(k)} - \mu \nabla_{\theta} \mathcal{L}(\theta^{(k)}), \quad k = 0, 1, 2, \dots, \quad (2.21)$$

where $\theta^{(k)}$ denotes the parameter vector after k steps. Rearranging (2.21) and dividing by the step size μ isolates the per-step increment as a difference quotient:

$$\frac{\theta^{(k+1)} - \theta^{(k)}}{\mu} = -\nabla_{\theta} \mathcal{L}(\theta^{(k)}). \quad (2.22)$$

The left-hand side of (2.22) already has the shape of a derivative: a difference of two parameter vectors divided by a spacing. It is not one yet. The index k counts iterations and carries no unit of time, so there is nothing for the difference to be taken with respect to. Reading (2.22) as the discretization of an ordinary differential equation therefore requires a new, continuous parameter along which the iterates can be laid out.

The idea is to regard the discrete sequence $\{\theta^{(k)}\}_{k=0}^{\infty}$ as samples of a smooth curve $\theta(\tau)$, read at a set of instants $\tau^{(k)}$,

$$\theta^{(k)} = \theta(\tau^{(k)}),$$

so that the recursion becomes a statement about $\theta(\tau)$ rather than about the index k . What remains is to fix the instants. The step size μ supplies their spacing: we assign to

the k -th iterate the training-time stamp

$$\tau^{(k)} = k\mu, \quad \text{so that} \quad \tau^{(k+1)} - \tau^{(k)} = \mu, \quad (2.23)$$

which places successive iterates μ apart along a time axis τ and gives $\theta^{(k+1)} = \theta(\tau^{(k)} + \mu)$. Substituting into the left-hand side of (2.22) recasts it as a forward finite-difference quotient of θ in τ :

$$\frac{\theta(\tau^{(k)} + \mu) - \theta(\tau^{(k)})}{\mu} = -\nabla_{\theta} \mathcal{L}(\theta(\tau^{(k)})). \quad (2.24)$$

As the step size $\mu \rightarrow 0$, the spacing in (2.23) shrinks to zero, the discrete stamps $\tau^{(k)}$ fill out a continuous variable τ , and the left-hand side of (2.24) converges to the derivative $d\theta/d\tau$ by definition of the derivative. The recursion thus becomes the gradient flow ODE:

$$\frac{d\theta}{d\tau} = -\nabla_{\theta} \mathcal{L}(\theta), \quad \theta(0) = \theta^{(0)}. \quad (2.25)$$

The gradient flow (2.25) is a central object in optimization, and many standard first-order methods can be recovered as different time-discretization schemes of this same continuous trajectory. The discrete update (2.21) is precisely the forward (explicit) Euler discretization of (2.25) with step size μ (SÜLI et al., 2003; BURDEN et al., 2011), so gradient descent is just one numerical integrator for the flow; applying instead the backward (implicit) Euler scheme (SÜLI et al., 2003) recovers the proximal gradient method (PARIKH et al., 2013). Reading optimizers as discretizations of (2.25) thus organizes them into a single family, and it is the continuous flow that we analyze in what follows.

2.3.3 Neural Tangent Kernel

The kernel derived in this subsection was introduced by Jacot et al. (2018) and carried to the physics-informed setting by Wang, Yu, et al. (2022); the derivation below follows their treatment, specialized to the data-fitting loss (2.20). Consider the MLP fit to the dataset $\{(x_i, u_i)\}_{i=1}^N$ of Section 2.3.1. Group the pointwise fitting errors into a single error vector:

$$e(\theta) = \begin{bmatrix} u_{\theta}(x_1) - u_1 \\ \vdots \\ u_{\theta}(x_N) - u_N \end{bmatrix} \in \mathbb{R}^N. \quad (2.26)$$

Up to the constant factor $1/N$, the loss (2.20) is the squared norm of this vector, which we use in the form

$$\mathcal{L}(\theta) = \frac{1}{2} \|e(\theta)\|^2 = \frac{1}{2} \sum_{i=1}^N e_i(\theta)^2, \quad (2.27)$$

where $e_i(\theta) = u_\theta(x_i) - u_i$ is the i -th component of the error vector (2.26). Define the Jacobian of the error vector with respect to the parameters:

$$J(\theta) = \frac{\partial e}{\partial \theta} \in \mathbb{R}^{N \times N_\theta}, \quad J_{ij} = \frac{\partial e_i}{\partial \theta_j}, \quad i = 1, \dots, N, \quad j = 1, \dots, N_\theta,$$

where N_θ is the total number of trainable parameters. Each row $J_i = (J_{i1}, \dots, J_{iN_\theta}) \in \mathbb{R}^{N_\theta}$ is the parameter-gradient of the i -th residual. Since $\mathcal{L}$ is the quadratic form (2.27), the j -th component of the loss gradient is:

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \sum_{i=1}^N e_i(\theta) \frac{\partial e_i}{\partial \theta_j} = \sum_{i=1}^N J_{ij} e_i(\theta) = (J(\theta)^T e(\theta))_j.$$

Collecting all $j = 1, \dots, N_\theta$ into a column vector:

$$\nabla_\theta \mathcal{L} = J(\theta)^T e(\theta) \in \mathbb{R}^{N_\theta}. \quad (2.28)$$

Substituting (2.28) into the gradient flow (2.25):

$$\frac{d\theta}{d\tau} = -J(\theta)^T e(\tau). \quad (2.29)$$

The error is written here as a function of τ rather than of θ . The two notations denote the same object: along the flow the parameters are themselves a function of training time, so $e(\tau) := e(\theta(\tau))$, and it is this composition whose evolution we now track. To find how the error vector itself evolves, differentiate each component $e_i(\theta(\tau))$ with respect to τ by the chain rule:

$$\frac{de_i}{d\tau} = \sum_{j=1}^{N_\theta} \frac{\partial e_i}{\partial \theta_j} \frac{d\theta_j}{d\tau} = J_i(\theta) \cdot \frac{d\theta}{d\tau}.$$

Stacking this identity for all $i = 1, \dots, N$ into a matrix equation:

$$\frac{de}{d\tau} = J(\theta) \frac{d\theta}{d\tau}.$$

Substituting (2.29):

$$\frac{de}{d\tau} = J(\theta) (-J(\theta)^T e(\tau)) = - \underbrace{J(\theta) J(\theta)^T}_{K(\theta) \in \mathbb{R}^{N \times N}} e(\tau). \quad (2.30)$$

The matrix $K(\theta) = J(\theta)J(\theta)^T$ is the empirical Neural Tangent Kernel (NTK) (JACOT et al., 2018). It is symmetric positive semi-definite by construction: for any $v \in \mathbb{R}^N$,

$$v^T K(\theta) v = v^T J(\theta) J(\theta)^T v = \|J(\theta)^T v\|^2 \geq 0.$$

The (i, j) entry of $K(\theta)$ follows directly from expanding the matrix product $J(\theta)J(\theta)^T$:

$$K_{ij}(\theta) = \sum_{l=1}^{N_\theta} J_{il}(\theta) J_{jl}(\theta) = \sum_{l=1}^{N_\theta} \frac{\partial e_i}{\partial \theta_l}(\theta) \frac{\partial e_j}{\partial \theta_l}(\theta) = \nabla_{\theta} e_i(\theta)^T \nabla_{\theta} e_j(\theta).$$

This is the inner product in $\mathbb{R}^{N_\theta}$ of the parameter gradients of the network output at the i -th and j -th training points, where $e_i = u_\theta(x_i) - u_i$. Hence $K(\theta) \in \mathbb{R}^{N \times N}$ is the Gram matrix of the network's sensitivities across the whole dataset, and equation (2.30) shows that it alone governs how the fitting error evolves under gradient flow.

2.3.4 Spectral Bias

The spectral bias (or frequency principle) is the empirical observation that neural networks trained by gradient-based methods reduce the low-frequency components of the error faster than the high-frequency ones (RAHAMAN et al., 2019). This section derives that phenomenon directly from the NTK dynamics established above. When the training points lie on a regular grid and the target function is periodic, the error admits a discrete Fourier representation, and one can ask how its frequency content is approximated along the training iterations. We first establish the decay in the eigenbasis of the NTK and then, by transforming to Fourier space, show that it is precisely the low frequencies that are resolved first.

For a network with a sufficiently large number of parameters, the Jacobian $J(\theta)$ changes negligibly during training, a regime known as lazy training (JACOT et al., 2018; CHIZAT et al., 2019; WANG; YU, et al., 2022). The motivation for it is a scaling argument. In the wide-network limit the output at any point is an aggregate of many hidden units, each contributing a vanishing share of it as the width grows. An $O(1)$ change in the function is then reached by many parameters each moving very little, instead of a few moving a lot. Since the Jacobian is itself a function of the parameters, it barely moves either, and in the infinite-width limit it stays frozen at its value at initialization (JACOT et al., 2018; CHIZAT et al., 2019). The networks trained here are finite, so this is an approximation, not an identity, and how far it holds is measured directly in Chapter 4. Writing $J_0 = J(\theta(0))$ and $K_0 = J_0 J_0^T$, the approximation $K(\theta(\tau)) \approx K_0$ turns

equation (2.30) into a linear ODE with constant coefficients:

$$\frac{de}{d\tau} = -K_0 e(\tau), \quad e(0) = e_0. \quad (2.31)$$

Written component-wise, the i -th row of this system is

$$\frac{de_i}{d\tau} = - \sum_{j=1}^N K_0^{ij} e_j(\tau), \quad i = 1, \dots, N,$$

where $K_0^{ij} = \nabla_{\theta_0} e_i^T \nabla_{\theta_0} e_j$ is the inner product, at initialization, of the parameter gradients of the network output at training points x_i and x_j . The rate of change of the error at point i is therefore a weighted sum of all current errors, with weights given by these pairwise kernel values. Through this coupling, the geometry of the sample grid and the architecture of the network jointly determine the convergence of every component.

Since $K_0 \in \mathbb{R}^{N \times N}$ is symmetric and positive semi-definite, the spectral theorem for symmetric matrices (STRANG, 2006; GOLUB et al., 2013) guarantees a decomposition

$$K_0 = V \Lambda V^T, \quad (2.32)$$

where $V = [v_1 \mid \dots \mid v_N]$ is an orthogonal matrix ($V^T V = I$) and

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N), \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq 0.$$

Introduce the change of coordinates

$$c(\tau) = V^T e(\tau), \quad (2.33)$$

so that, since V is orthogonal, $e(\tau) = V c(\tau)$. Differentiating (2.33) with respect to τ and substituting the ODE (2.31):

$$\frac{dc}{d\tau} = V^T \frac{de}{d\tau} = -V^T K_0 e(\tau).$$

Inserting $K_0 = V \Lambda V^T$ and $e(\tau) = V c(\tau)$:

$$\frac{dc}{d\tau} = -V^T (V \Lambda V^T) (V c(\tau)) = -(V^T V) \Lambda (V^T V) c(\tau) = -\Lambda c(\tau),$$

using $V^T V = I$ at each step. Since Λ is diagonal, this decouples into N independent scalar equations:

$$\frac{dc_k}{d\tau} = -\lambda_k c_k(\tau), \quad k = 1, \dots, N. \quad (2.34)$$

Each equation in (2.34) is a first-order linear ODE with constant coefficient (BOYCE et al., 2012). Separating the variables and integrating both sides from the initial state at $\tau = 0$ up to time τ , with s and r as integration variables for the amplitude and the time,

$$\int_{c_k(0)}^{c_k(\tau)} \frac{ds}{s} = -\lambda_k \int_0^\tau dr \quad \implies \quad \ln \frac{|c_k(\tau)|}{|c_k(0)|} = -\lambda_k \tau.$$

Exponentiating both sides yields the solution

$$c_k(\tau) = c_k(0) e^{-\lambda_k \tau}, \quad (2.35)$$

where $c_k(0) = v_k^T e_0$ is the projection of the initial error e_0 onto the k -th eigenvector of K_0 . For $\lambda_k = 0$ the equation gives $c_k(\tau) = c_k(0)$ for all τ : that eigendirection never decays.

Returning to the original coordinates via $e(\tau) = Vc(\tau)$:

$$e(\tau) = \sum_{k=1}^N (v_k^T e_0) e^{-\lambda_k \tau} v_k. \quad (2.36)$$

The error decomposes into N independent eigendirections. Each eigenvector v_k carries initial amplitude $v_k^T e_0$ and decays exponentially at rate λ_k : eigenvectors with large λ_k are suppressed quickly, while those with small λ_k persist throughout training. This per-eigendirection decay, each at the rate set by its eigenvalue, is the rigorous statement of spectral bias established by Cao et al. (2021); the same spectral ordering governs generalization as the training set grows (BORDELON et al., 2020).

Equation (2.36) is the continuous flow; gradient descent takes finite steps, and its error evolution inherits the same eigenstructure in discrete form. The link between the two is the training-time stamp of (2.23): iterate n sits at continuous time $\tau^{(n)} = n\mu$, so the iteration count and the training time are read off one another through

$$\tau = n\mu, \quad n = \tau/\mu.$$

Section 2.3.2 read the update (2.21) as forward Euler applied to the gradient flow in the parameters. The same statement is needed for the error, since that is what the analysis follows. There are two ways to obtain it: discretize the parameter update and then linearize, or linearize first, into the error equation (2.31), and discretize that. Both give the same law.

Take the first. Gradient descent on the loss is

$$\theta^{(n+1)} = \theta^{(n)} - \mu \nabla_{\theta} \mathcal{L}(\theta^{(n)}).$$

Under lazy training the Jacobian is frozen at J_0 , so (2.28) gives $\nabla_{\theta}\mathcal{L}(\theta^{(n)}) = J_0^T e^{(n)}$ and the step is

$$\Delta\theta^{(n)} := \theta^{(n+1)} - \theta^{(n)} = -\mu J_0^T e^{(n)}.$$

Expanding the error to first order in that displacement,

$$e^{(n+1)} = e^{(n)} + J_0 \Delta\theta^{(n)} + O(\|\Delta\theta^{(n)}\|^2),$$

and substituting the step,

$$e^{(n+1)} = e^{(n)} - \mu J_0 J_0^T e^{(n)}.$$

Now the second. Equation (2.31) is already linear in e . The forward, or explicit, Euler scheme (SÜLI et al., 2003; LEVEQUE, 2007) replaces its derivative at $\tau^{(n)}$ by a forward difference over one step of length μ ,

$$\left. \frac{de}{d\tau} \right|_{\tau^{(n)}} \approx \frac{e^{(n+1)} - e^{(n)}}{\mu},$$

and evaluates the right-hand side at the current iterate,

$$\frac{e^{(n+1)} - e^{(n)}}{\mu} = -K_0 e^{(n)}.$$

Multiplying through by μ ,

$$e^{(n+1)} = e^{(n)} - \mu K_0 e^{(n)}.$$

Since $K_0 = J_0 J_0^T$, the two results are the same equation. Collecting terms gives the single-step law

$$e^{(n+1)} = (I - \mu K_0) e^{(n)}. \quad (2.37)$$

One gradient-descent step of size μ is one forward-Euler step of length μ on the error. Iterating from $e^{(0)} = e_0$ and diagonalizing with $K_0 = V\Lambda V^T$,

$$e^{(n)} = (I - \mu K_0)^n e_0 = \sum_{k=1}^N (v_k^T e_0) (1 - \mu \lambda_k)^n v_k. \quad (2.38)$$

The continuous solution reaches the same expression, one mode at a time. Evaluate the decay factor of (2.36) at $\tau = n\mu$ and split it across the n steps,

$$e^{-\lambda_k \tau} = e^{-\lambda_k n\mu} = (e^{-\lambda_k \mu})^n,$$

then expand the per-step factor and truncate at first order in $\mu\lambda_k$ (BURDEN et al., 2011),

$$e^{-\lambda_k\mu} = 1 - \mu\lambda_k + O((\mu\lambda_k)^2) \approx 1 - \mu\lambda_k. \quad (2.39)$$

Substituting term by term carries (2.36) into (2.38), so $(1 - \mu\lambda_k)^n$ is the forward-Euler counterpart of $e^{-\lambda_k\tau}$. The two agree as the step vanishes with $\tau = n\mu$ held fixed,

$$\lim_{\mu \rightarrow 0} (1 - \mu\lambda_k)^{\tau/\mu} = e^{-\lambda_k\tau}.$$

The same truncation constrains the step. Under the continuous flow every mode with $\lambda_k > 0$ decays for any $\mu > 0$. The discrete mode contracts only when its amplification factor is smaller than one in magnitude,

$$|1 - \mu\lambda_k| < 1 \quad \Longleftrightarrow \quad 0 < \mu\lambda_k < 2,$$

which is the absolute-stability condition of the explicit Euler method (LEVEQUE, 2007; SÜLI et al., 2003). The product $\mu\lambda_k$ alone fixes how the k -th mode behaves, and it does so in four regimes. Below one, the factor lies in $(0, 1)$ and the mode decays monotonically toward zero; at exactly one it vanishes and the mode is removed in a single step. Between one and two the factor is negative, so the mode still decays but flips sign at every step. Past two it exceeds one in magnitude and the mode grows without bound. A single step size therefore acts differently on every eigendirection of K_0 .

Since $\lambda_1 = \lambda_{\max}$ is the largest eigenvalue, every mode stays stable only when the largest one does, which caps the learning rate at

$$\mu < \frac{2}{\lambda_{\max}}.$$

The minimal experiment of Section 3.5 sets its step from this limit: rather than a nominal value, μ is fixed at a constant fraction of $2/\lambda_{\max}$ at every width, small enough to keep every mode in the monotone regime $\mu\lambda_k < 1$ while staying comparable across widths whose $\lambda_{\max}$ differ.

The same argument gives the time each mode needs. From (2.35), the amplitude of the k -th eigendirection at time τ is

$$|c_k(\tau)| = |v_k^T e_0| e^{-\lambda_k\tau}.$$

Require it to fall below a target precision $\varepsilon > 0$,

$$|v_k^T e_0| e^{-\lambda_k\tau} < \varepsilon,$$

and divide by $|v_k^T e_0| > 0$,

$$e^{-\lambda_k \tau} < \frac{\varepsilon}{|v_k^T e_0|}.$$

Taking logarithms preserves the inequality,

$$-\lambda_k \tau < \ln \frac{\varepsilon}{|v_k^T e_0|},$$

and dividing by $-\lambda_k < 0$ reverses it,

$$\tau > \frac{1}{\lambda_k} \ln \frac{|v_k^T e_0|}{\varepsilon}.$$

The minimum training time to resolve the k -th eigendirection to precision ε is therefore

$$\tau_k = \frac{1}{\lambda_k} \ln \frac{|v_k^T e_0|}{\varepsilon}. \quad (2.40)$$

The projection enters (2.40) through a logarithm and the eigenvalue through $1/\lambda_k$, so λ_k sets the scale of τ_k while the projection only shifts it. Since $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq 0$, the times are ordered $\tau_1 \leq \tau_2 \leq \dots \leq \tau_N$: higher-indexed eigendirections require more training time, in inverse proportion to their eigenvalue. Arora et al. (2019) reached similar conclusions from a discrete-time analysis of gradient descent, showing that the convergence speed of each eigenvector is governed by its eigenvalue and by the projection of the target onto that eigenvector.

The decomposition (2.36) is expressed in the eigenbasis $\{v_k\}_{k=1}^N$ of the NTK, which need not coincide with any physically meaningful basis. When the N training points lie on a regular grid and the target is periodic, however, the natural basis is the Fourier one, and recasting the dynamics there shows that the decay is genuinely organized by frequency. Let $F \in \mathbb{C}^{N \times N}$ be the Fourier matrix, that is, the discrete Fourier transform (DFT) matrix with entries

$$F_{jl} = \frac{1}{\sqrt{N}} \omega^{jl}, \quad \omega = e^{-2\pi i/N}, \quad j, l = 0, \dots, N-1.$$

With this normalization F is unitary, $F^* F = I$, so that $F^{-1} = F^*$, where F^* is the conjugate transpose (GOLUB et al., 2013; STRANG, 2006). Applying F to the error vector yields its spectrum, with inverse transform F^{-1} :

$$\hat{e}(\tau) = F e(\tau), \quad e(\tau) = F^{-1} \hat{e}(\tau) = F^* \hat{e}(\tau). \quad (2.41)$$

In particular $\hat{e}_0 = F e_0$ is the spectrum of the initial error, the difference between the

spectra of the initial network output and the target. Since F is constant in τ , applying it to the linearized dynamics (2.31) carries the error evolution into the frequency domain:

$$\frac{d\hat{e}}{d\tau} = F \frac{de}{d\tau} = -FK_0 e(\tau) = -FK_0 F^{-1} \hat{e}(\tau).$$

Inserting the eigendecomposition $K_0 = V\Lambda V^T$ from (2.32),

$$FK_0 F^{-1} = FV\Lambda V^T F^{-1} = (FV)\Lambda(FV)^* = (FV)\Lambda(FV)^{-1},$$

where $V^T F^{-1} = V^* F^* = (FV)^*$ uses that V is real orthogonal, and $(FV)^* = (FV)^{-1}$ uses that a product of unitary matrices is unitary. The columns of FV , namely the Fourier transforms $\{Fv_k\}_{k=1}^N$ of the NTK eigenvectors, therefore diagonalize the frequency-domain dynamics. Defining the modal amplitudes $\tilde{c}(\tau) = (FV)^{-1}\hat{e}(\tau)$ and repeating the steps (2.33)–(2.35) gives

$$\frac{d\tilde{c}_k}{d\tau} = -\lambda_k \tilde{c}_k(\tau) \quad \implies \quad \tilde{c}_k(\tau) = \tilde{c}_k(0) e^{-\lambda_k \tau}. \quad (2.42)$$

These are the same amplitudes as before: $\tilde{c} = (FV)^{-1}Fe = V^{-1}e = V^T e = c$, the projection of the error onto the NTK eigenvectors, now read in Fourier space. Transforming back through (2.41) yields the evolution of the error spectrum,

$$\hat{e}(\tau) = \sum_{k=1}^N [(Fv_k)^* \hat{e}_0] e^{-\lambda_k \tau} (Fv_k). \quad (2.43)$$

Equation (2.43) is the spectral-bias statement in the literal sense of the frequency spectrum: the error spectrum is a superposition of the transformed eigenvectors Fv_k , each damped at its own rate λ_k . The frequencies resolved first are those carried by the eigenvectors with the largest eigenvalues; empirically these eigenvectors are smooth and concentrated in the low Fourier modes (RAHAMAN et al., 2019; XU et al., 2020), so the low-frequency content of the error decays first while the high-frequency content persists.

To close, we specialize the initial condition. The weights of an MLP are drawn at random from a distribution centred at zero, so at small initialization scale $\theta_0 \approx 0$ and the network output u_{θ_0} is close to zero across the whole grid. Set against a target u of order one, it is negligible. This is a common simplification in spectral-bias analyses, and under

it the initial error is the target with opposite sign, as is its spectrum:

$$\hat{e}_0 = F(u_{\theta_0} - u) = \hat{u}_{\theta_0} - \hat{u} \approx -\hat{u} = - \begin{bmatrix} \hat{u}(0) \\ \vdots \\ \hat{u}(N-1) \end{bmatrix}. \quad (2.44)$$

Substituting (2.44) into (2.43) casts the dynamics as a reconstruction of the target frequency by frequency: each Fourier mode of u is acquired at a rate set by the NTK eigenvalues, the smooth low-frequency content emerging first and the fine high-frequency detail last. This is the regime anticipated at the start of the section, a periodic target on a regular grid, in which the quality of the approximation can be tracked mode by mode along the training iterations. It is also the setting we implement: the minimal experiment of Chapter 3 fits a single Boussinesq elevation profile on exactly such a grid, so that (2.44) holds by construction and the mode-by-mode reconstruction can be measured against the prediction.

This last observation can be made quantitative. Taking the modulus of the modal amplitude (2.42) isolates the magnitude of the k -th component along training,

$$|\tilde{c}_k(\tau)| = |\tilde{c}_k(0)| e^{-\lambda_k \tau}, \quad |\tilde{c}_k(0)| = |(Fv_k)^* \hat{e}_0| \approx |(Fv_k)^* \hat{u}|, \quad (2.45)$$

where the last estimate uses the small-scale initialization (2.44), so that $|(Fv_k)^* \hat{u}|$ measures how strongly the target overlaps with the k -th eigen-mode. Training, however, advances in discrete steps: by the construction of the gradient flow (2.23), the continuous time and the iteration count n are related by $\tau = n\mu$, with μ the learning rate. Substituting $\tau = n\mu$ into (2.45) and requiring the amplitude to fall below a target precision $\varepsilon > 0$,

$$|(Fv_k)^* \hat{u}| e^{-\lambda_k n\mu} < \varepsilon \implies n > \frac{1}{\mu \lambda_k} \ln \frac{|(Fv_k)^* \hat{u}|}{\varepsilon},$$

gives an estimate of the number of gradient-descent iterations needed to resolve the k -th frequency:

$$n_k = \left\lceil \frac{1}{\mu \lambda_k} \ln \frac{|(Fv_k)^* \hat{u}|}{\varepsilon} \right\rceil. \quad (2.46)$$

The count follows just as directly from the discrete recursion, without passing through the continuous flow. The discrete counterpart of (2.45) is the modal amplitude read off (2.38), $|\tilde{c}_k^{(n)}| = |\tilde{c}_k(0)| |1 - \mu\lambda_k|^n$, and requiring it below ε gives

$$|\tilde{c}_k(0)| |1 - \mu\lambda_k|^n < \varepsilon \implies n > \frac{1}{-\ln |1 - \mu\lambda_k|} \ln \frac{|\tilde{c}_k(0)|}{\varepsilon}, \quad (2.47)$$

the inequality flipping because in the stable regime $|1 - \mu\lambda_k| < 1$, so $\ln |1 - \mu\lambda_k| < 0$.

This exact discrete count reduces to the continuous estimate (2.46): by the first-order truncation (2.39), $-\ln|1 - \mu\lambda_k| = \mu\lambda_k + O((\mu\lambda_k)^2)$, so the denominator collapses to $\mu\lambda_k$ and, with $|\tilde{c}_k(0)| \approx |(Fv_k)^*\hat{u}|$, equation (2.47) becomes (2.46). Substituting $\tau = n\mu$ into the continuous decay and counting steps directly on the discrete recursion therefore agree to leading order in the step, exactly as the forward-Euler link between (2.36) and (2.38) requires.

This estimate is tested numerically in this work. The minimal experiment of Chapter 3 records, for each eigendirection, the iteration at which the measured amplitude first falls below ε , and sets it against the count predicted by (2.47). The discrete form is the one put to the test, since it rests on the same geometric law $|1 - \mu\lambda_k|^n$ as the measured decay; the continuous estimate (2.46) would add the truncation of the exponential to whatever discrepancy the measurement shows. Because (2.47) carries no reference to the width of the network, it predicts that networks of different widths fall on a single curve, which Chapter 4 checks at three widths.

2.3.5 Physics-Informed Neural Networks

Introduced by Raissi et al. (2019), Physics-Informed Neural Networks (PINNs) insert the PDE directly into the optimization process, forcing the neural network to satisfy the model equations and approximating the PDE solution.

Consider the general PDE Initial Value Problem:

$$\begin{cases} \mathcal{D}[u](x, t) = 0, & \forall (x, t) \in \Omega \times (0, T], \\ u(x, 0) = u_0(x), & x \in \Omega, \end{cases} \quad (2.48)$$

where $\mathcal{D}$ is a differential operator on u , $\Omega \times (0, T]$ is the spatio-temporal domain, and u_0 is the initial condition.

The PDE residual loss penalizes violation of the governing equation at collocation points $\{(x_i, t_i)\}_{i=1}^{N_D} \subset \Omega \times (0, T]$:

$$\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{N_D} \sum_{i=1}^{N_D} \|\mathcal{D}[u_\theta](x_i, t_i)\|^2.$$

The initial condition is enforced at collocation points $\{x_i\}_{i=1}^{N_0} \subset \Omega$:

$$\mathcal{L}_{\text{ic}}(\theta) = \frac{1}{N_0} \sum_{i=1}^{N_0} \|u_\theta(x_i, 0) - u_0(x_i)\|^2.$$

The combined PINN loss is:

$$\mathcal{L}_{\text{PINN}}(\theta) = \mathcal{L}_{\text{pde}}(\theta) + \mathcal{L}_{\text{ic}}(\theta). \quad (2.49)$$

Due to the compositional structure of MLPs, PINN implementations evaluate PDE residuals via Automatic Differentiation (AD) (LINNAINMAA, 1970; PASZKE et al., 2019), using Adam (KINGMA et al., 2014) or L-BFGS (NOCEDAL, 1980) as the optimizer.

Application to the Boussinesq Dataset

For the Boussinesq system, a separate PINN is trained for each sample $(\alpha_i, \beta_i) \in \mathcal{S}$ (2.19). The differential operator $\mathcal{D}$ in the generic problem (2.48) is instantiated as the pair of Boussinesq residuals. Each acts on both predicted fields and carries the coefficients of the sample, so we write them with all four arguments displayed:

$$\begin{aligned} \mathcal{D}_1[\eta_\theta, u_\theta, \alpha_i, \beta_i] &= (\eta_\theta)_t + (u_\theta)_x + \alpha_i (\eta_\theta u_\theta)_x, \\ \mathcal{D}_2[\eta_\theta, u_\theta, \alpha_i, \beta_i] &= (u_\theta)_t - \frac{\beta_i}{3}(u_\theta)_{xxt} + (\eta_\theta)_x + \alpha_i u_\theta (u_\theta)_x. \end{aligned}$$

The combined loss (2.49) for sample i then reads:

$$\begin{aligned} \mathcal{L}_{\text{PINN}}^i(\theta) &= \frac{1}{N_{\mathcal{D}}} \sum_{l=1}^{N_{\mathcal{D}}} \left[\mathcal{D}_1[\eta_\theta, u_\theta, \alpha_i, \beta_i](x_l, t_l)^2 + \mathcal{D}_2[\eta_\theta, u_\theta, \alpha_i, \beta_i](x_l, t_l)^2 \right] \\ &\quad + \frac{1}{N_0} \sum_{l=1}^{N_0} \left[(\eta_\theta(x_l, 0) - \eta_0(x_l))^2 + (u_\theta(x_l, 0) - u_0(x_l))^2 \right], \end{aligned} \quad (2.50)$$

where η_0 and u_0 are given by the shared initial condition (2.2). Because the parameters (α_i, β_i) enter the residuals explicitly, each PINN encodes knowledge of a single sample and must be retrained for every new parameter pair.

2.3.6 Spectral Bias in Physics-Informed Neural Networks

The Neural Tangent Kernel analysis of Sections 2.3.3–2.3.4 was carried out for an MLP fitting data on a uniform grid, the setting in which the kernel is approximately circulant and its eigenvectors are genuine Fourier modes. A PINN does not fit data directly: its loss (2.49) combines an initial-condition term with a PDE-residual term. The same machinery nevertheless applies once both constraints are stacked into a single error vector, now indexed over the N_r residual points in place of the N data points of the

MLP,

$$e(\theta) = \begin{bmatrix} u_\theta(x_1, 0) - u_0(x_1) \\ \vdots \\ u_\theta(x_{N_0}, 0) - u_0(x_{N_0}) \\ \mathcal{D}[u_\theta](x_1, t_1) \\ \vdots \\ \mathcal{D}[u_\theta](x_{N_D}, t_{N_D}) \end{bmatrix} \in \mathbb{R}^{N_r}, \quad N_r = N_0 + N_D, \quad (2.51)$$

so that, up to normalization, $\mathcal{L}_{\text{PINN}}$ reduces to the same quadratic form $\frac{1}{2}\|e\|^2$ analyzed for the MLP. With this definition the gradient-flow derivation leading to (2.30) carries over verbatim, with the Jacobian now also containing the parameter gradients of the PDE residuals, and the error still obeys the linear dynamics $\frac{de}{d\tau} = -K_0 e$, with convergence of each eigenvector governed by its eigenvalue exactly as in (2.40) (WANG; YU, et al., 2022).

This prediction is borne out empirically. Spectral bias is a frequently reported obstacle in PINN training: networks fit smooth, low-frequency solutions readily but stall on sharp gradients and high-frequency content (RAHAMAN et al., 2019; XU et al., 2020; WANG; TENG, et al., 2021), and the same pathology underlies several documented PINN failure modes (KRISHNAPRIYAN et al., 2021). Wang, Yu, et al. (2022) traced the effect directly to the NTK, showing that its spectrum is concentrated at low eigenvalues, so that high-frequency PDE residuals are the last to be minimized. For nonlinear wave equations of Boussinesq type, Wang et al. (2024) report the same difficulty in resolving the finer wave structure; the two-field Boussinesq system studied here has not been examined in this respect, which is part of what Chapter 4 sets out to do. A range of remedies target this imbalance directly, among them Fourier feature mappings that recondition the kernel spectrum (TANCIK et al., 2020), adaptive loss weighting (WANG; TENG, et al., 2021), and second-order or spectrum-aware optimizers (KHODAKARAMI et al., 2026). Pursuing these mitigations is beyond the scope of this work: our aim is to apply the SciML methods to the Boussinesq system and to expose and understand the spectral bias that emerges there, not to remove it. Understanding the effect nonetheless remains essential background for the neural-operator methods studied in the remainder of this work, whose ability to capture the high-frequency content of the Boussinesq solutions is examined directly in Chapter 4.

2.4 Neural Operators

Neural operators generalize neural networks from learning finite-dimensional mappings to learning operators between infinite-dimensional function spaces. This section introduces the general framework, then the Fourier Neural Operator (FNO) as the principal

architecture, and finally the Physics-Informed Neural Operator (PINO) as the physics-constrained extension.

2.4.1 General Definition

Let $D \subset \mathbb{R}^d$ be a bounded open domain and let $\mathcal{A}(D; \mathbb{R}^{d_a})$ and $\mathcal{U}(D; \mathbb{R}^{d_u})$ be Banach spaces of functions (KREYSZIG, 1978). The target operator is a (potentially nonlinear) map

$$G^\dagger : \mathcal{A} \rightarrow \mathcal{U}, \quad a \mapsto u = G^\dagger(a),$$

where both the input a and the output u are functions defined on D .

In this work, $G^\dagger$ is the solution operator of the Boussinesq system: given an input a encoding the PDE coefficients (α, β) together with the initial condition (2.2), the operator returns the full space-time solution (η, u) . Because the initial condition is the same for every sample of the dataset (2.19), it carries nothing that distinguishes one sample from another, and the input reduces to the pair (α_i, β_i) , which is how the operator is written from here on.

A neural operator G_θ approximates $G^\dagger$ by alternating linear layers with pointwise nonlinearities, in the composition

$$G_\theta = Q \circ \sigma(W_L + \mathcal{K}_L) \circ \cdots \circ \sigma(W_1 + \mathcal{K}_1) \circ P, \quad (2.52)$$

where:

- $P : \mathbb{R}^{d_a} \rightarrow \mathbb{R}^{d_c}$ is the lifting operator, mapping the input $a(x)$ pointwise to a higher-dimensional channel representation with $d_c \gg d_a$;
- $\sigma(W_\ell + \mathcal{K}_\ell)$, for $\ell = 1, \dots, L$, are the operator layers, each combining a local linear map W_ℓ with a nonlocal integral operator $\mathcal{K}_\ell$, followed by a pointwise activation σ ;
- $Q : \mathbb{R}^{d_c} \rightarrow \mathbb{R}^{d_u}$ is the projection operator mapping the final channel representation back to the output dimension.

What makes (2.52) an operator, rather than a map between finite-dimensional vectors, is the nonlocal term $\mathcal{K}_\ell$ in each layer. It is an integral operator,

$$(\mathcal{K}_\ell v)(x) = \int_D \kappa_\ell(x, y) v(y) dy,$$

where $\kappa_\ell : D \times D \rightarrow \mathbb{R}^{d_c \times d_c}$ is a learned kernel. Each output point therefore draws on the whole input function, and not only on its value at x , which is what the local map

W_ℓ alone would give. Different choices of kernel parametrization for each $\mathcal{K}_\ell$ give rise to different neural operator architectures.

2.4.2 Fourier Neural Operators

Presented by Li et al. (2021), the Fourier Neural Operator (FNO) is the neural operator (2.52) in which each $\mathcal{K}_\ell$ uses a shift-invariant kernel: $\kappa_\ell(x, y) = \kappa_\ell(x - y)$. This assumption turns each integral operator into a convolution, which by the convolution theorem (DEBNATH, 2012) can be evaluated efficiently in the frequency domain.

Each spectral layer updates the channel representation via:

$$v_{\ell+1}(x) = \sigma\left((W_\ell v_\ell)(x) + (\mathcal{K}_\ell v_\ell)(x)\right). \quad (2.53)$$

The operator $\mathcal{K}_\ell$ acts via convolution with the shift-invariant kernel κ_ℓ :

$$(\mathcal{K}_\ell v_\ell)(x) = \int_D \kappa_\ell(x - y) v_\ell(y) dy.$$

By the convolution theorem (DEBNATH, 2012), this equals:

$$(\mathcal{K}_\ell v_\ell)(x) = \mathcal{F}^{-1}\left(\mathcal{F}(\kappa_\ell) \cdot \mathcal{F}(v_\ell)\right)(x). \quad (2.54)$$

Property (2.54) is what makes the layer (2.53) affordable: the nonlocal integral never has to be formed in physical space, since a forward transform, a pointwise multiplication and an inverse transform deliver it. To see how this is parametrized, recall that for a periodic function f on $[0, 2\pi]$:

$$f(x) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{ikx}, \quad \hat{f}(k) = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-ikx} dx.$$

The convolution $(\kappa * v)$ simplifies to:

$$(\kappa * v)(x) = \int_0^{2\pi} \kappa(x - y) v(y) dy = \sum_{k \in \mathbb{Z}} (2\pi \hat{\kappa}(k) \hat{v}(k)) e^{ikx}.$$

Instead of learning κ_ℓ as a function in physical space, the FNO directly parametrizes its Fourier coefficients as learnable complex-valued matrices $R_{\ell,k} \approx 2\pi \hat{\kappa}_\ell(k)$. Only the $|k| \leq k_{\max}$ low-frequency modes are retained, with all others set to zero; $k_{\max}$ is a hyperparameter.

Training Objective

The FNO is trained as a pure supervised learning problem. Given the dataset $\mathcal{S}$ (2.19), whose i -th element pairs the coefficients (α_i, β_i) with the reference solution (η_i, u_i) computed for them, the training objective minimizes the mean squared error between the predicted and reference space-time fields:

$$\mathcal{L}_{\text{FNO}}(\theta) = \frac{1}{M} \sum_{i=1}^M \frac{1}{N_x N_t} \sum_{j=0}^{N_x-1} \sum_{n=0}^{N_t} \left[(\eta_{\theta}^i(x_j, t_n) - \eta_i(x_j, t_n))^2 + (u_{\theta}^i(x_j, t_n) - u_i(x_j, t_n))^2 \right], \quad (2.55)$$

where G_{θ} is the neural operator (2.52) and $(\eta_{\theta}^i, u_{\theta}^i) := G_{\theta}(\alpha_i, \beta_i)$ is its prediction for sample i . Minimizing (2.55) therefore drives $G_{\theta}(\alpha_i, \beta_i)$ towards (η_i, u_i) for all M samples of $\mathcal{S}$ at once, which is what lets a single trained operator answer the whole parameter family.

Application to Non-Time-Periodic Problems

When the solution is also assumed to be periodic in time with period T (so that $D = [0, 2\pi] \times [0, T]$), shift-invariance extends to the time direction as well: $\kappa_{\ell}(x, y, t, \tau) = \kappa_{\ell}(x - y, t - \tau)$. The convolution then generalizes to:

$$(\mathcal{K}_{\ell} v_{\ell})(x, t) = \int_D \kappa_{\ell}(x - y, t - \tau) v_{\ell}(y, \tau) dy d\tau = \mathcal{F}^{-1} \left(\mathcal{F}(\kappa_{\ell}) \cdot \mathcal{F}(v_{\ell}) \right)(x, t),$$

and the FNO parametrizes the two-dimensional Fourier modes (k_x, k_t) with $|k_x| \leq k_{x,\max}$ and $|k_t| \leq k_{t,\max}$, both hyperparameters.

For problems that are not time-periodic, however, applying the Fourier transform along the temporal axis implicitly extends the signal periodically, potentially introducing a jump discontinuity at the temporal boundary. This can trigger the Gibbs phenomenon (GIBBS, 1899), producing oscillatory artifacts near the boundary that persist regardless of how many temporal modes are retained. Standard signal-processing techniques such as zero-padding and window functions (e.g. Hann, Hamming windows) can partially mitigate these effects, at the cost of altering the temporal structure of the signal. Li et al. (2021) acknowledge this limitation and provide an alternative formulation in which the Fourier layers operate only along the spatial axis while the operator advances forward in time, avoiding the periodicity assumption in the temporal direction entirely. In this work we deliberately retain the two-dimensional space-time formulation, in which the Fourier layers act on both the spatial and the temporal axes, so that a single operator emits the entire trajectory in one forward pass. This keeps the architecture simple and uniform across space and time, but it reintroduces the temporal-periodicity assumption for a Boussinesq solution that is not time-periodic, and therefore admits a Gibbs artifact at the temporal

boundary $t = T$. Rather than suppress this artifact with windowing, we retain the plain formulation and examine the boundary behaviour directly in Chapter 4, where it turns out to distinguish the purely data-driven operator from its physics-informed variant.

2.4.3 Physics-Informed Neural Operators

Analogously to how PINNs add a physics-informed residual loss to the MLP training objective, the Physics-Informed Neural Operator (PINO) (LI et al., 2023a) incorporates PDE constraints into the FNO training. The FNO backbone remains unchanged; what differs is the loss function, which now combines a data term with residuals evaluated on the predicted space-time fields.

Unlike a PINN, which represents the solution as a single continuous function and differentiates it via automatic differentiation, the FNO outputs (η_θ, u_θ) as a discrete $N_x \times N_t$ array in a single forward pass over the full space-time domain. Evaluating the PDE residuals therefore reduces to numerically differentiating this output array.

Spatial derivatives are computed pseudospectrally using ξ_k (2.13): the DFT of η_θ along the spatial axis at each time t_n yields $\hat{\eta}_\theta(k, t_n)$, from which

$$(\eta_x)_j = \mathcal{F}^{-1}(i \xi_k \hat{\eta}_\theta(k, t_n))_j, \quad (\eta_{xx})_j = \mathcal{F}^{-1}(-\xi_k^2 \hat{\eta}_\theta(k, t_n))_j,$$

and identically for u_θ . Time derivatives are computed by finite differences: second-order central in the interior and first-order one-sided at the boundaries,

$$(\eta_\theta)_t^n = \begin{cases} \frac{\eta_\theta^{n+1} - \eta_\theta^n}{\Delta t}, & n = 0, \\ \frac{\eta_\theta^{n+1} - \eta_\theta^{n-1}}{2 \Delta t}, & 1 \leq n \leq N_t - 1, \\ \frac{\eta_\theta^n - \eta_\theta^{n-1}}{\Delta t}, & n = N_t, \end{cases}$$

and identically for $(u_\theta)_t^n$. The mixed term $(u_\theta)_{xxt}$ is obtained by first computing $(u_\theta)_{xx}$ spectrally and then applying the time-difference stencil to the result. This combination of spectral spatial derivatives and finite-difference temporal derivatives follows the reference implementation of Li et al. (2023a) (LI et al., 2023b).

With all derivatives available, the Boussinesq equations (2.1) define two differential operators acting on the predicted fields:

$$\mathcal{D}_1(\eta_\theta, u_\theta) := (\eta_\theta)_t + (u_\theta)_x + \alpha (\eta_\theta u_\theta)_x, \quad (2.56)$$

$$\mathcal{D}_2(\eta_\theta, u_\theta) := (u_\theta)_t - \frac{\beta}{3} (u_\theta)_{xxt} + (\eta_\theta)_x + \alpha (u_\theta (u_\theta)_x). \quad (2.57)$$

Here $(\eta_\theta^i, u_\theta^i) := G_\theta(\alpha_i, \beta_i)$ denotes the network output for sample i , and the subscript $_{j,n}$ indicates evaluation at the grid point (x_j, t_n) . The PINO loss is a weighted sum of three terms:

$$\mathcal{L}_{\text{PINO}}(\theta) = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}}(\theta) + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}}(\theta) + \lambda_{\text{data}} \mathcal{L}_{\text{data}}(\theta), \quad (2.58)$$

where:

- $\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{M} \sum_{i=1}^M \frac{1}{N_x N_t} \sum_{j=0}^{N_x-1} \sum_{n=0}^{N_t} \left[\mathcal{D}_1(\eta_\theta^i, u_\theta^i)_{j,n}^2 + \mathcal{D}_2(\eta_\theta^i, u_\theta^i)_{j,n}^2 \right]$ penalizes violation of the Boussinesq equations at each grid point, with residuals evaluated using the parameters (α_i, β_i) of each sample;
- $\mathcal{L}_{\text{ic}}(\theta) = \frac{1}{M} \sum_{i=1}^M \frac{1}{N_x} \sum_{j=0}^{N_x-1} \left[(\eta_\theta(x_j, 0) - \eta_0(x_j))^2 + (u_\theta(x_j, 0) - u_0(x_j))^2 \right]$ enforces the shared initial condition (2.2);
- $\mathcal{L}_{\text{data}}(\theta) = \frac{1}{M} \sum_{i=1}^M \frac{1}{N_x N_t} \sum_{j=0}^{N_x-1} \sum_{n=0}^{N_t} \left[(\eta_\theta^i(x_j, t_n) - \eta_i(x_j, t_n))^2 + (u_\theta^i(x_j, t_n) - u_i(x_j, t_n))^2 \right]$ measures deviation from the reference solutions $(\eta_i, u_i) \in \mathcal{S}$ (2.19);
- $\lambda_{\text{pde}}, \lambda_{\text{ic}}, \lambda_{\text{data}} > 0$ are weighting hyperparameters.

The PINO therefore generalizes the FNO by adding physics supervision: it can be trained with fewer labeled samples by relying on the PDE residual for additional guidance.

3 METHODOLOGY

This chapter shows how the experiments were built, on what data, and under which rules, in enough detail to reproduce them. Three things are common to every experiment, and we settle them first: the computational setup (Section 3.1), the pseudospectral dataset (Section 3.2), and the error metrics that score every method on one scale (Section 3.3). Section 3.4 then defines the six trained methods. Section 3.5 sets up the experiment on spectral bias, which runs on the same Boussinesq solution and the same kind of network as the rest of the work, but with time and parameters held fixed, and Section 3.6 condenses the hyperparameters of this work in one table.

3.1 Computational Setup and Reproducibility

The software behind these experiments was all written for this work. It runs on Python 3.11 with PyTorch 2.12.0 (PASZKE et al., 2019) built against CUDA 13.0, on a single NVIDIA GeForce GTX 1660 SUPER GPU. The static figures were drawn with Plotly and the animations with Matplotlib. We used no pre-trained weights, no third-party operator-learning library, and no external dataset. The pseudospectral solver, the four architectures, and each optimization loop are our own code, which let us instrument them for the spectral analysis and the custom visualization used in upcoming chapters. The full implementation is available at <https://github.com/samuelkutz/TCC>.

Each learned architecture stays close to its original implementation. The Fourier Neural Operator follows Li et al. (2021); the Physics-Informed Neural Operator follows Li et al. (2023a), together with its released code (LI et al., 2023b); and the Physics-Informed Neural Network follows Raissi et al. (2019). All of them are written from scratch in PyTorch, and where an implementation differs from its reference we point that out.

Reproducibility rests on two conventions. A single global seed initializes Python, NumPy and PyTorch at the start of the pipeline and puts the cuDNN backend in deterministic mode. The stages then run in a fixed order, each inheriting the random state from its predecessors.

We now fix the experimental parameters shared by every method. Starting with

the physical model, the spatial and temporal domains of the Boussinesq system (2.1) are

$$x \in [-L, L], \quad L = 60, \quad t \in [0, T], \quad T = 30. \quad (3.1)$$

Every experiment uses the initial condition (2.2) at unit amplitude $A = 1$, that is, $\eta(x, 0) = \text{sech}^2(x)$ and $u(x, 0) = 0$. A unit crest sits at the centre of a domain 120 units wide. That width is deliberate: the two counter-propagating pulses of the wave-equation limit (2.9)–(2.10) travel at unit speed, so over the horizon $T = 30$ they cover half the distance to the imposed periodic boundary and never reach it. Table 3.1 lists the constants that hold everywhere. Each of them is restated below at the point where it first matters.

Table 3.1 – Fixed simulation, dataset, and evaluation constants shared by every experiment. Unless a section says otherwise, these values hold throughout.

Parameter	Value
<i>Domain and initial condition</i>	
Spatial half-length L ($\Omega = [-L, L]$)	60
Final time T ($t \in [0, T]$)	30
Initial amplitude A	1
Initial condition	$\eta(x, 0) = \text{sech}^2(x)$, $u(x, 0) = 0$
<i>Dataset solve</i>	
Spatial grid points N_x	128
Time levels (127 RK4 steps)	128
Spatial spacing $\Delta x = 2L/N_x$	0.9375
Temporal spacing $\Delta t = T/127$	≈ 0.2362
<i>Parameter sampling</i>	
Coupled coefficients	$\alpha = \beta$
Training samples M	20
Sampling grid	<code>linspace(0.1, 4.0, 20)</code>
<i>Evaluation and reproducibility</i>	
Evaluation values $\alpha = \beta$	{0.1, 3.21, 4.2}
Evaluation resolutions N_x	{128, 256, 512}
Spectral snapshot resolution	256
Global seed	37

Source: The author (2026).

3.2 The Parametric Dataset

The reference solutions are the supervised targets that the data-driven methods learn. This role asks for a high-fidelity baseline, and the pseudospectral Runge–Kutta solver of Section 2.2 supplies one since the domain is wide enough that the periodicity the solver assumes is not a problem.

We keep the parameter family one-dimensional on purpose. Following Section 2.2.4, the initial conditions are fixed and only the PDE coefficients vary, tied together as the pair $\{(\alpha_i = \beta_i), (\eta_i, u_i)\}$. For small values of the parameters, the dynamics stay close to linear. Each value in (??) is handed to the solver on $N_x = 128$ spatial points and marched through 127 Runge–Kutta steps over $[0, T]$, giving $N_t = 128$ time levels. The grid spacings are

$$\Delta x = \frac{2L}{N_x} = \frac{120}{128} = 0.9375, \quad \Delta t = \frac{T}{127} = \frac{30}{127} \approx 0.2362. \quad (3.2)$$

A solve returns the two space-time solution matrices $\eta_i(x_j, t_n)$ and $u_i(x_j, t_n)$ on the $N_x \times N_t$ grid, and we collect them into the solution dataset $\mathcal{S}$ of Equation (2.19). Each sample is stored with four input channels and two output channels. The inputs are the initial conditions $\eta_i(x_j, 0)$ and $u_i(x_j, 0)$ at every spatial point x_j , together with the two coefficients α_i and β_i (summing four feature channels), each of the four held constant along time so that all of them fill the space-time grid:

$$X \in \mathbb{R}^{M \times 4 \times N_x \times N_t}, \quad Y \in \mathbb{R}^{M \times 2 \times N_x \times N_t}. \quad (3.3)$$

The two output channels carry η_i and u_i themselves.

From now on the panels report only the surface elevation η , since it carries the wave structure under study and the velocity should be similar.

The quantities a network reads here do not share a scale. The fields η and u are of order one, the coefficients α and β span multiple values, and the coordinates span $[-L, L] \times [0, T]$. Following standard practice, we need a way to standardize the data. For pointwise networks the recommendation is to non-dimensionalize, scaling the variables so that they become dimensionless and of order one, since disparate scales let one variable dominate another in the gradient and force the optimizer into small steps for one and large steps for the next (WANG; SANKARAN, et al., 2023); for operators, the reference implementations normalize each channel by an affine map as a matter of course (KOSSAIFI et al., 2024; LI et al., 2021).

We use a single rule for all of it. Writing $[f_{\min}, f_{\max}]$ for the envelope of a quantity f , every network input in this work is carried onto $[-1, 1]$ by

$$\tilde{f} = \frac{2(f - f_{\min})}{f_{\max} - f_{\min}} - 1, \quad f = f_{\min} + \frac{1}{2}(\tilde{f} + 1)(f_{\max} - f_{\min}), \quad (3.4)$$

the second expression being the exact inverse of the first. The target $[-1, 1]$, not $[0, 1]$, suits the hyperbolic tangent of the pointwise backbones, which is odd and centred at the origin, and it is the interval the minimal experiment of Section 3.5 already works on.

Where each method uses it

The MLP applies (3.4) to its two coordinates and leaves its output in physical units. The PINN does the same, and applies it inside the forward pass; the residual is then differentiated with respect to the physical x and t , automatic differentiation carries the chain rule through the rescaling on its own, and what is enforced stays the residual of (2.1) and not of a rescaled equation. Applying the same map to both also leaves the two networks sharing an input scale as well as a backbone, so the comparison between them isolates the training signal.

The FNO and the PINO apply (3.4) to the four input channels and the two output channels of (3.3). To these four inputs both append two coordinate channels, x and t on the same $[-1, 1]$, so the lifting layer reads six channels instead of four; the spatial one follows the solver’s periodic grid, whose right endpoint is dropped, so that the channel keeps a fixed physical meaning as the resolution changes. The two operators differ in the units each loss term is formed in. Both compare prediction against target in the normalized variables, which puts the pure-data and data-and-physics regimes on one objective scale. The PINO then returns its fields to physical units for the residual and initial-condition terms, so that the constraint it enforces is the balance of (2.1) at its true magnitudes. Every prediction is returned to physical units by the inverse in (3.4) before any reported metric is formed.

3.3 Error Metrics

A comparison is only fair if every method is measured against the same reference and on the same scales. Each method is scored against a freshly recomputed pseudospectral reference at three values of the parameter of (2.1),

$$\alpha = \beta \in \{0.1, 3.21, 4.2\}, \quad (3.5)$$

the first near-linear and itself a sample of the training grid (??), the second more nonlinear and falling between two training samples, the third more nonlinear still and past the upper end of the sampling range. The middle value 3.21 is the default wherever a single representative parameter is needed, as it is for the MLP and the two PINNs, which are trained at one parameter.

To test the discretization invariance that operators are supposed to enjoy, each trained operator is queried without retraining on grids of resolution

$$N_x \in \{128, 256, 512\}, \quad (3.6)$$

the training resolution and its two- and fourfold refinements, all powers of two, the sizes on which the fast Fourier transform of Section 2.2.1 runs most efficiently (TREFETHEN, 2000). For each query the input channels of Section 3.2 are rebuilt on the finer grid from the same initial condition, and the reference is recomputed at the matching resolution, so prediction and reference are always compared at the same points. The pointwise methods are queried on the same grids, since a continuous map can be sampled anywhere.

Let η_{true} and η_{pred} be the reference and predicted elevation on a shared space-time grid. We report three complementary numbers.

Time-resolved relative error

At each time level t_n we take the spatial L^2 error and divide it by the norm of the reference at that instant,

$$\mathbb{E}(t_n) = \frac{\|\eta_{\text{pred}}(\cdot, t_n) - \eta_{\text{true}}(\cdot, t_n)\|_2}{\|\eta_{\text{true}}(\cdot, t_n)\|_2}. \quad (3.7)$$

The quotient is defined at every time level without regularization: the reference is a travelling wave of fixed amplitude on a periodic domain, so $\|\eta_{\text{true}}(\cdot, t_n)\|_2$ stays of order one along the whole horizon and never approaches zero. Read against time, $\mathbb{E}(t_n)$ shows where along the trajectory the error piles up, and its spread over all time levels is summarized by a box plot. This is the quantity behind the parameter and resolution panels of Chapter 4.

Absolute spectral error

To see which spatial frequencies are wrong, we take the spatial real DFT at a fixed time and compare amplitudes mode by mode. Writing $\hat{\eta}(k, t_n)$ for the DFT (2.11) of the elevation at wavenumber index k , the per-mode error is the absolute difference of amplitudes,

$$\mathbb{E}_{\text{spec}}(k, t_n) = \frac{1}{N_x} \left| |\hat{\eta}_{\text{pred}}(k, t_n)| - |\hat{\eta}_{\text{true}}(k, t_n)| \right|. \quad (3.8)$$

The factor $1/N_x$ undoes the scaling the unnormalized transform (2.11) carries with the grid size, so the amplitudes, and with them the error, are read in the units of η and mean the same thing on each of the grids (3.6). Measuring the difference of amplitudes rather than their ratio is what keeps the metric honest across the spectrum. A reference spectrum does not decay smoothly: it passes through near-nulls, single modes whose amplitude falls orders of magnitude below their neighbours, and dividing by one of those turns an unremarkable amplitude error into a spike that reports on the reference rather than on the method. Any relative form has to be defended against that with a floor under

the denominator, and the floor then becomes a free parameter that decides which modes are reported faithfully. The absolute error carries no denominator and needs no such choice: every mode contributes what it is worth in amplitude, so the modes that carry the solution dominate the reading and the near-nulls, which carry nothing, contribute nothing. What the measure gives up is the built-in reference a ratio provides, so the panels of Chapter 4 print $\mathbb{E}_{\text{spec}}$ beside the reference spectrum on the same logarithmic decades: a mode is reproduced while its error stays well below the amplitude the reference carries there, and it is lost where the two meet.

Frequency-band error during training

To watch spectral bias form as the optimizer runs, we split the spatial spectrum into three bands and save each band error every 500 optimizer iterations. The per-mode quantity is exactly the absolute spectral error (3.8), and it is reduced to one number per band in two averages. First over the N_t time levels, giving a value for each mode, then over the modes of the band,

$$\overline{\mathbb{E}}_{\text{spec}}(k) = \frac{1}{N_t} \sum_{n=1}^{N_t} \mathbb{E}_{\text{spec}}(k, t_n), \quad \mathbb{E}_{\text{band}}(B) = \frac{1}{|B|} \sum_{k \in B} \overline{\mathbb{E}}_{\text{spec}}(k), \quad (3.9)$$

so the same metric drives the per-frequency spectra and the band-tracking curves. With $N_k = \lfloor N_x/2 \rfloor + 1$ retained real-DFT modes, and writing $q_1 = \lfloor N_k/4 \rfloor$ and $q_2 = \lfloor N_k/2 \rfloor$ for the two cut points, the three index sets B are

$$B_{\text{low}} = \{k : 0 \leq k < q_1\}, \quad B_{\text{mid}} = \{k : q_1 \leq k < q_2\}, \quad B_{\text{high}} = \{k : q_2 \leq k < N_k\}, \quad (3.10)$$

a quarter of the range each for the low and mid bands and the upper half for the high band. These curves feed the band-tracking comparison of Chapter 4. Every method is tracked at one common parameter, $\alpha = \beta = 3.21$: the pointwise MLP and PINN against their own training solution at that value, each operator against the dataset sample nearest it, so the bands are read at a single parameter across all six.

3.4 The Six Methods

Four architectures give rise to six trained methods, arranged along two axes. The first is the architecture family. The pointwise networks, the multilayer perceptron of Section 2.3.1 and the physics-informed network of Section 2.3.5, represent one solution as a continuous map of the coordinates, $(x, t) \mapsto \eta$ for the MLP and $(x, t) \mapsto (\eta, u)$ for the PINN, and are retrained for each parameter. The operator networks, the Fourier neural

operator of Section 2.4.2 and its physics-informed variant of Section 2.4.3, learn the whole parameter family (??) at once and return a new parameter’s solution in a single forward pass.

The second axis is the training regime, that is, what the loss is allowed to see:

- Pure data fits the reference solutions with a supervised term alone and no equation. The data-only MLP and the FNO sit here.
- Data and physics keeps that supervised term and adds the Boussinesq residual beside it. The PINN and the PINO with data sit here.
- Pure physics switches the supervised term off, leaving the equation residual and the initial condition to drive the network. The PINN and the PINO without data sit here.

The MLP and the FNO carry no residual, so each family contributes a single pure-data method; the PINN and the PINO carry a residual whose data weight can be set on or off, so each contributes one data-and-physics and one pure-physics method. Together these give the 2×3 methods collected in Table 3.2.

Every method is trained against the pseudospectral references of Section 3.2, on the grid of Table 3.1, and is scored by the same metrics of Section 3.3, so a difference in behaviour traces to the family and the regime rather than to the setup. The operators read all $M = 20$ samples of (3.3) at once, while each pointwise method is tied to a single parameter and fits the one solve at $\alpha = \beta = 3.21$.

3.5 Spectral-Bias Setting

To some degree every method in this work carries a version of the spectral bias derived in Section 2.3.4. We examine it on the simplest object this work can offer, and we keep it on the Boussinesq system with a multilayer perceptron from $\mathbb{R}$ to $\mathbb{R}$ that learns a single elevation profile at a fixed time $t = 15$, taken from the same pseudospectral solver at the intermediate parameter $\alpha = \beta = 3.21$,

$$x \longmapsto \eta_\theta(x) \approx \eta(x, 15), \quad x \in [-L, L]. \quad (3.11)$$

Nothing is introduced that the rest of the work does not already use: no auxiliary equation, no synthetic target, no second architecture. This leaves the setting the theory of Section 2.3.4 was derived for, a target sampled on a regular periodic grid and fitted by gradient descent.

Fixing the time removes the temporal axis from the problem, so the Fourier analysis of the error is a plain spatial transform on the $N_x = 128$ grid and the wavenumbers of

the theory are the wavenumbers of the figure, with no averaging in between. The instant $t = 15$ sits at the middle of the horizon, by which point the initial crest has split into two counter-propagating pulses with dispersive oscillations between them, so the target carries amplitude across the spectrum rather than concentrating it in the single smooth bump the initial condition would offer.

3.5.1 Setting Up the Experiment

We train three simple networks, at widths $N_{\text{MLP}} \in \{8, 64, 512\}$, each with four hidden layers and hyperbolic-tangent activations, one input and one output. The same standardization map (3.4) is then used so these networks read the same scales as the rest of the work; the figures undo the target scaling to display η in physical units. Every figure in this section comes from those same three runs.

Because the target lives on a single grid of $N_x = 128$ points, the Jacobian $J \in \mathbb{R}^{N_x \times N_\theta}$ is cheap enough to be fully computed, one backward pass per grid point, at every width including the widest. That matters, because the empirical kernel

$$K_0 = J_0 J_0^\top = V \Lambda V^\top \quad (3.12)$$

has to be eigendecomposed and it is only 128×128 . It is also cheap enough to rebuild the kernel during training, which lets us see how far it drifts from its initial state, an important check.

Training is full-batch gradient descent on the sum-of-squares loss $\frac{1}{2}\|e\|^2$, so that a single step is exactly

$$\theta \leftarrow \theta - \mu J^\top e, \quad (3.13)$$

the update behind (2.37), with $e = \eta_\theta - \eta_{\text{true}}$ measured on the grid.

Each width takes the same fixed fraction of its own stability bound, $\mu = 0.2 (2/\lambda_{\text{max}}) = 0.4/\lambda_{\text{max}}$, with λ_{max} the largest eigenvalue of K_0 at that width. Since λ_{max} grows with width, the three step sizes differ; what they share is the product $\mu\lambda_{\text{max}} = 0.4$, which places all three networks at the same point inside the monotone regime $\mu\lambda_k < 1$ of Section 2.3.4.

Anchoring μ to the bound $\mu < 2/\lambda_{\text{max}}$ instead of to a nominal value keeps the three widths comparable. We iterate 500,000 times.

The MLP is nonlinear in its parameters, so its Jacobian is not constant: J drifts away from J_0 as training proceeds, and the initial-kernel prediction $e^{(n)} = (I - \mu K_0)^n e_0$ of (2.38), which comes from the forward-Euler discretization of the gradient flow (2.31), is an approximation here. Whether it survives contact with a real network is part of what the experiment measures.

We therefore record four measurements along the runs and set each against the

initial-kernel prediction of Section 2.3.4.

- Per-eigendirection error decay. We project the measured error onto each eigenvector v_k of K_0 , giving $c_k^{(n)} = v_k^\top e^{(n)}$, and compare its amplitude against the predicted $|c_k(0)| |1 - \mu\lambda_k|^n$ of (2.38).
- Per-mode spectral decay. We take the real DFT of the error profile at each iteration, giving one amplitude per wavenumber, and compare the measured decay against the prediction of (2.43). Four fixed wavenumbers are followed across the whole run rather than four eigendirections, so the panel reads the frequency principle directly.
- Band-averaged absolute spectral error. The same comparison over the whole spectrum instead of four sampled wavenumbers: the per-mode error (3.8) is averaged within each of the three bands of (3.10). There is a single time level here, so the time average of (3.9) does not apply. Measured and predicted fields are both scored against the reference profile, which puts this panel on the metric the six trained methods are tracked with in Chapter 4.
- Per-mode iteration count n_k . For each eigendirection we record the iteration at which $|c_k^{(n)}|$ first drops below a precision ε and compare it against the discrete crossing time $n_k = \ln(|c_k(0)|/\varepsilon)/(-\ln |1 - \mu\lambda_k|)$ of (2.47), the step at which $|1 - \mu\lambda_k|^n$ of (2.38) falls below the selected error ε . We set $\varepsilon = 0.05 \max_k |c_k(0)|$, a fraction of the largest initial coefficient rather than an absolute number, so the threshold scales with the amplitude of η instead of with the units it is measured in.

The first three panels are drawn at the widest network, since lazy training is an assumption about wide networks; the fourth collects all three widths. Alongside these we keep what the networks actually produced: the final output at each width, and the widest network's output at logarithmically spaced checkpoints, both set against the reference profile.

3.5.2 Kernel Drift and the Eigenvalue Spectrum

The predictions above assume the kernel stays put, and we measure whether it does along the same three runs. We log the relative parameter drift $\|\theta - \theta_0\|/\|\theta_0\|$ and the relative kernel drift $\|K - K_0\|/\|K_0\|$ at logarithmically spaced iterations, and we record the eigenvalue spectrum of K at initialization and again at the end of training.

The two drift curves ask whether the kernel stays put, as lazy training assumes, and whether it does so more convincingly as the network widens, as the initial-kernel approximation (2.31) predicts. The spectrum exposes the eigenvalue disparity that (2.40) names as the cause of spectral bias. Together with the four checks of Section 3.5.1, these

curves open the results of Chapter 4, where they anchor the reading of the spectral-bias curves measured on the full architectures.

3.6 Configuration Summary

Table 3.2 gathers the configuration of every experiment. The four learned architectures share a common budget, 15,000 iterations of Adam at learning rate 10^{-3} , so the comparison of Chapter 4 reflects the architecture and the training regime, not the effort spent on tuning. The last row reports the measured training wall-clock on the hardware of Section 3.1, which is indicative and depends on the GPU used.

Table 3.2 – Configuration of the four training architectures and of the minimal spectral-bias network. The FNO and PINO share the operator backbone and its parameter count; the MLP and PINN share the feed-forward backbone and are single-parameter methods. All Adam-trained methods use 15,000 iterations. The PINN draws a fresh set of 5,000 interior collocation points at every iteration and evaluates its initial-condition term on a further 500 points. The last column is a separate network, deliberately minimal, $\mathbb{R} \rightarrow \mathbb{R}$ on the single profile $\eta(x, 15)$, trained once per width.

Setting	MLP	PINN	FNO	PINO	Minimal MLP
Input $\rightarrow$ output	$\mathbb{R}^2 \rightarrow \mathbb{R}$	$\mathbb{R}^2 \rightarrow \mathbb{R}^2$	operator	operator	$\mathbb{R} \rightarrow \mathbb{R}$
Trainable parameters	198,401	198,658	2,106,146	2,106,146	241, 12,673, 789,505
Hidden layers	4	4	4 Fourier	4 Fourier	4
Width / channels	256	256	32	32	8, 64, 512
Fourier modes (per axis)	—	—	16	16	—
Optimizer	Adam	Adam	Adam	Adam	GD
Learning rate	10^{-3}	10^{-3}	10^{-3}	10^{-3}	$0.4/\lambda_{\max}$
Iterations	15,000	15,000	15,000	15,000	500,000
Parameters trained	single (3.21)	single (3.21)	all 20	all 20	single (3.21)
Collocation / batch	full batch	5,000 pts	batch 256	batch 256	128 pts (full)
Loss weights pde/ic/data	— / — / 1	1 / 1 / {1, 0}	— / — / 1	100 / 1 / {1, 0}	— / — / 1
Physics-residual grid	—	5,000 pts	—	256×256	—
Target	$\eta(x, t)$	η, u	family	family	$\eta(x, 15)$
Data regimes	data only	with / without	data only	with / without	data only
Training wall-clock	≈ 152 s	≈ 880 s	≈ 1178 s	≈ 4440 – 5460 s	—

Source: The author (2026).

4 NUMERICAL RESULTS

This chapter reports what the experiments of Chapter 3 produced. Section 4.1 accounts for what each method costs to train. The next three sections then take the methods along one axis at a time with Section 4.2 sweeping through model parameters at a fixed resolution, Section 4.3 holding the parameter fixed and querying the operators off their training grid to check zero-shot superresolution, and Section 4.4 fixing both and comparing the spectra mode by mode at three time instants. The temporal artifact given by the Gibbs phenomenon that the physics term corrects is isolated in Section 4.5. Section 4.6 aims to establish spectral bias with the initial-kernel analysis of a minimal MLP fitting the Boussinesq wave profile, and Section 4.7 follows the error through the frequency bands as training proceeds, showing the phenomenon exists for all methods. Section ?? gathers all of our findings. The pseudospectral solver is the ground truth throughout.

4.1 Training Cost

Before asking how well the methods perform, we ask what they cost to train. Table 4.1 lists the size, wall-clock training time, and final training loss of the six trained methods. It is important to notice that neural operator methods carry many more trainable parameters than the neural network methods. Specifically, PINO is slower because it assembles its physics residual on the finer 256×256 grid (Table 3.2) at every iteration.

Table 4.1 – Architecture size, training time on the GTX 1660 SUPER, and final training loss. The final-loss column is not comparable across rows. Accuracy is measured instead by the relative-error metrics of the following sections.

Method	Regime	Parameters	Train time (s)	Final loss
MLP	pure data	198,401	151.9	4.3×10^{-5}
FNO	pure data	2,106,146	1177.8	4.8×10^{-7}
PINO	data and physics	2,106,146	5459.4	5.2×10^{-5}
PINO	pure physics	2,106,146	4441.2	4.0×10^{-5}
PINN	data and physics	198,658	898.9	7.1×10^{-5}
PINN	pure physics	198,658	868.4	4.6×10^{-5}

Source: The author (2026).

4.2 Across the Parameter Range

The first comparison holds the resolution fixed and sweeps the parameters $\alpha = \beta$. It concerns the operators alone, since each pointwise MLP and PINN is trained at a single parameter and has nothing to say about the sweep.

The box plots at the foot of Figures 4.1 and 4.2 summarize the time-resolved error (3.7) over the whole trajectory at each of the three parameters, and the two operators order them differently. The FNO is worst at the near-linear extreme and eases as the nonlinearity grows; the PINO with data traces a shallow U across the axis, easiest at the middle parameter.

In the PINO the physics residual holds the low and mid modes across the range, leaving what error remains at the extremes, where the dynamics are hardest. The panels agree only that the error is non-monotone in the parameter; which end is worst depends on the method. They differ in that the FNO curves end in a sharp terminal spike and the PINO curves do not. Section 4.5 takes that up.

4.3 Across Resolutions

The next comparison queries the operator off its training resolution, probing the discretization invariance operators are meant to enjoy, as noted earlier. Figure 4.3 takes the PINO with data, whose supervised term saw the dataset at resolution 128 while its physics residual was formed on a 256×256 grid (Table 3.2), and evaluates it at 128, 256, and 512 at the fixed parameter $\alpha = \beta = 3.21$ with no retraining.

Discretization invariance holds in the geometric sense. At every resolution the operator returns a smooth, coherent surface with the correct wave structure, with no retraining and no architectural change, since the input is simply resampled on the finer grid.

Queried below that resolution the operator loses the residual-resolved detail; queried above it, it meets high-wavenumber content it never learned to produce, since its spectral layers keep only a fixed number of Fourier modes per axis. The plain FNO, whose only constraint is the data at its training resolution, degrades monotonically over the same sweep, best on the grid it was trained on and worse on every finer one (Figure 4.4).

4.4 Reproducing the Spectrum

To compare the families on equal terms we fix $\alpha = \beta = 3.21$ and inspect the full spatial spectrum of the prediction at three times, $t = 0$, $t = T/2 = 15$, and $t = T = 30$. Figures 4.5–4.10 report the six methods. In every panel the black curve is the reference spatial spectrum and the dashed orange curve is the prediction, with the per-

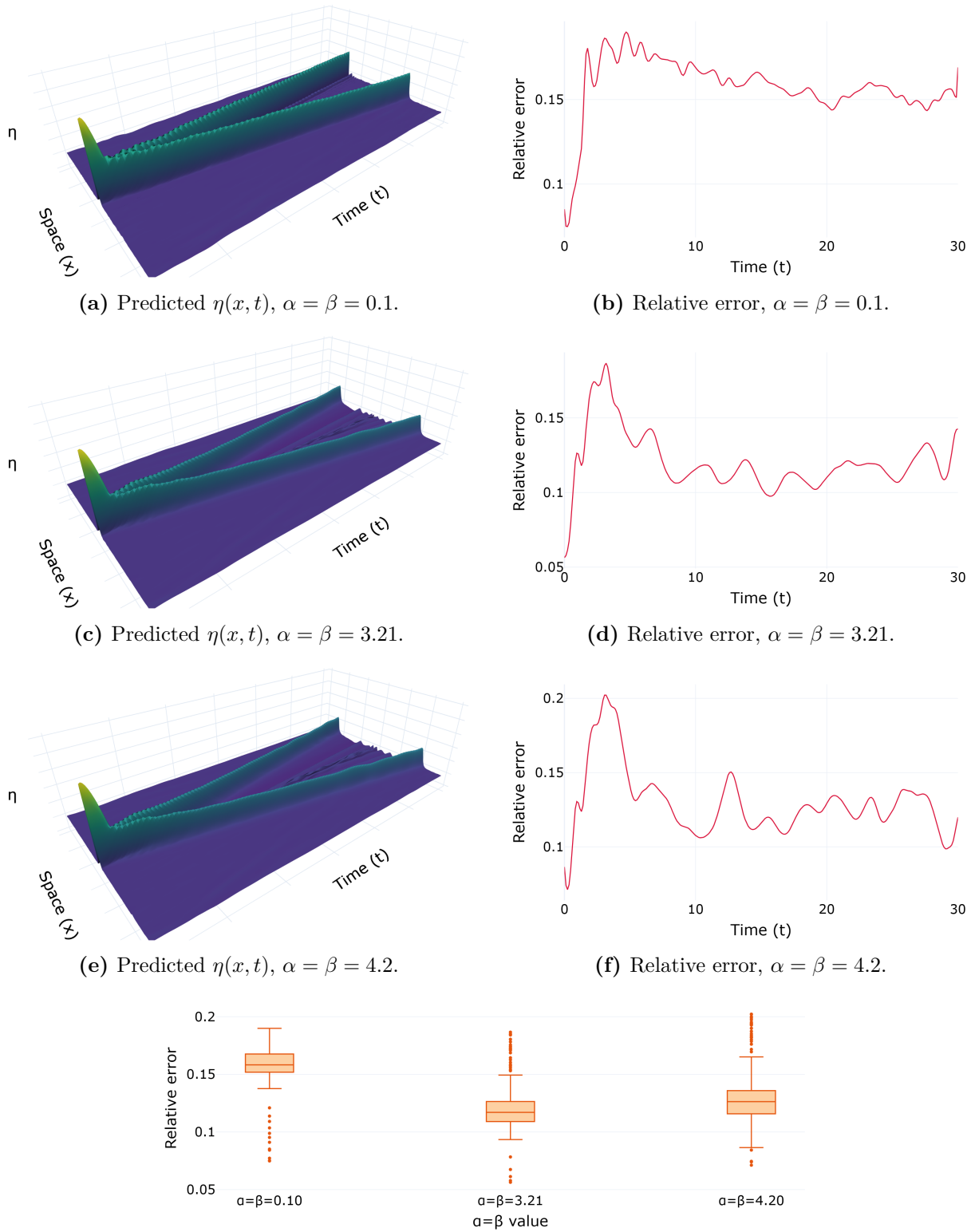


Figure 4.1 – FNO (pure data) across the nonlinearity axis ($\alpha = \beta \in \{0.1, 3.21, 4.2\}$), evaluated at resolution 256. The predicted surfaces (left) capture the two counter-propagating pulses, but the time-resolved relative error (right) spikes at the final time $t = 30$, the temporal Gibbs artifact of treating time as a periodic axis.

Source: The author (2026).

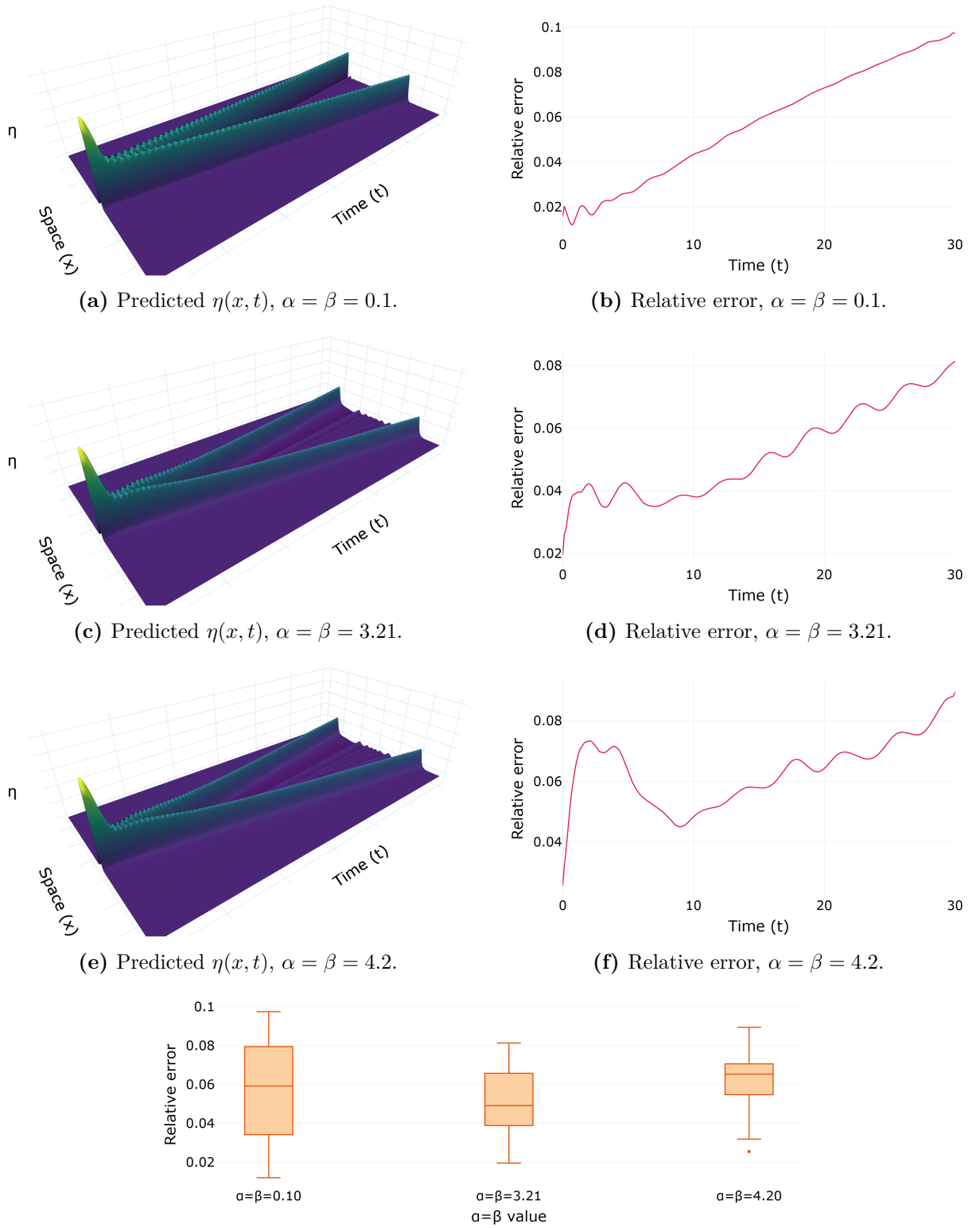


Figure 4.2 – PINO (data and physics) across the same nonlinearity axis and resolution. The predicted surfaces are visually indistinguishable from the FNO, but the terminal-time spike of Figure 4.1 is gone: the error grows gently across the trajectory instead of jumping at the boundary. The physics residual and the initial-condition term act as a temporal regularizer.

Source: The author (2026).

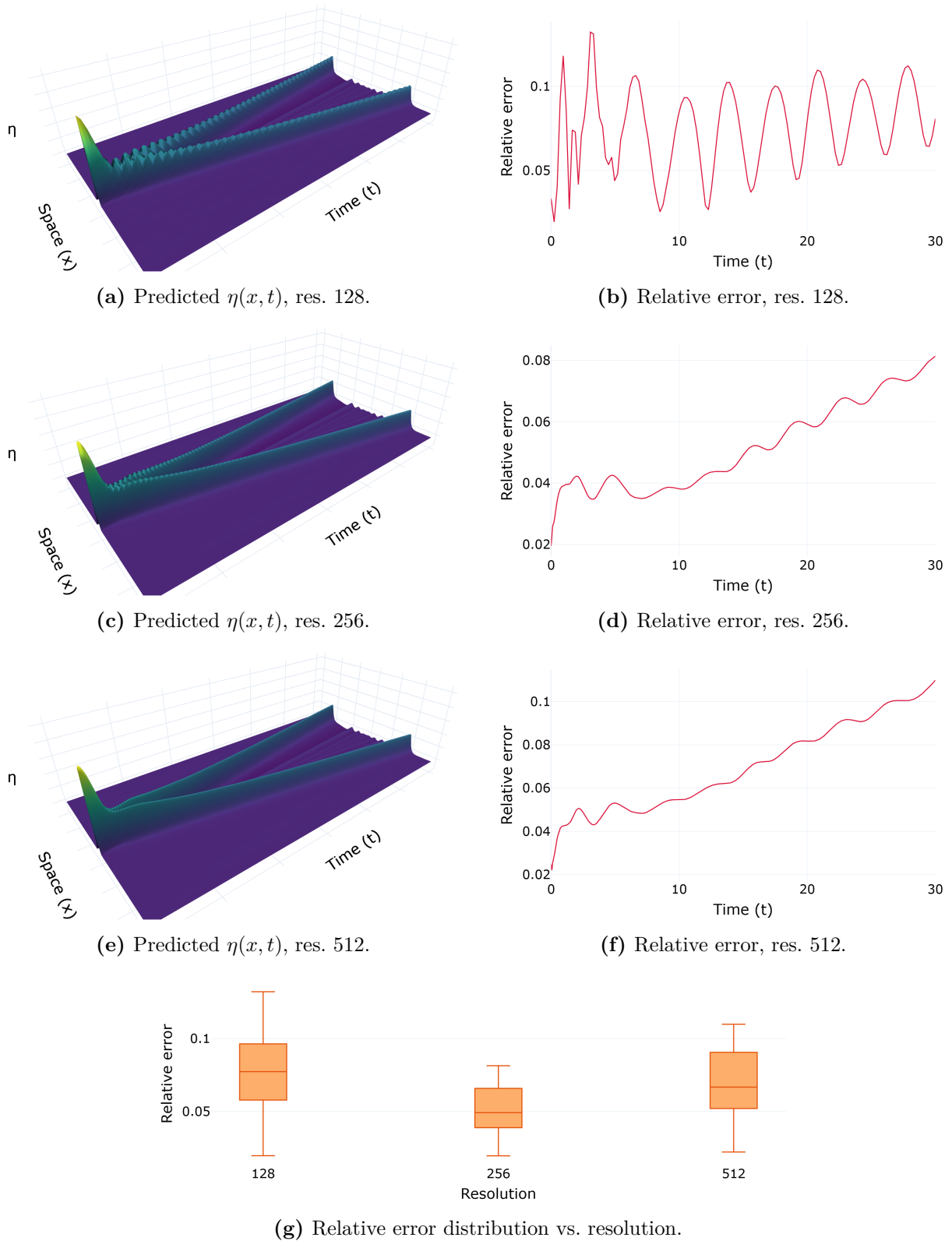


Figure 4.3 – PINO (data and physics) queried zero-shot at resolutions 128, 256 and 512 at $\alpha = \beta = 3.21$. The operator returns a coherent field at every resolution, and the relative error against the pseudospectral reference is lowest at the resolution on which the physics residual was trained and rises on either side of it, most on the coarsest grid: accuracy is anchored to the residual grid, not the training-data grid.

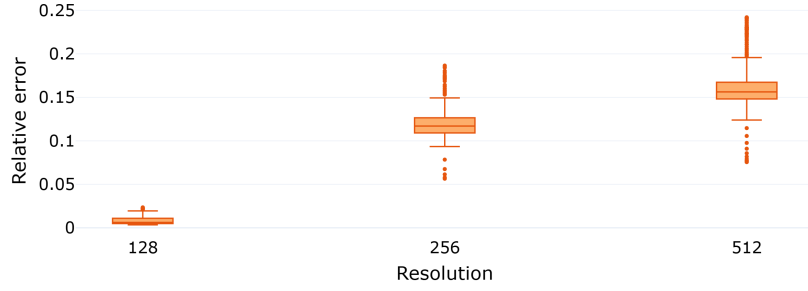


Figure 4.4 – Plain FNO queried zero-shot on the same three grids at $\alpha = \beta = 3.21$. With no physics residual to anchor it, the relative error is lowest on the training grid and grows with every refinement, in contrast to the interior minimum of Figure 4.3.

Source: The author (2026).

mode absolute error beside each spectrum and the mode-averaged error against time at the foot of each figure. One reading note applies throughout. The per-mode error (3.8) is a difference of amplitudes in the units of η , not a ratio, so it is read against the reference amplitude at the same wavenumber.

The MLP (Figure 4.5) fits the reference where the frequencies are low and undershoots where they are high, with no equation and no physics term in the objective to complicate the reading. Both PINN regimes fail to represent the high-wavenumber content of the sech^2 profile at $t = 0$. Without data (Figure 4.6) the predicted spectrum sheds most of its mid- and high-frequency amplitude by late time; adding data (Figure 4.7) tightens the low-wavenumber crests, in step with the low-band gain of Figure 4.14.

The operators, learning in the frequency domain, invert this picture at low and mid wavenumbers. The FNO (Figure 4.8) tracks the low and mid spectrum closely at early times, because those are the modes its spectral layers parametrize and learn. The PINO with data (Figure 4.9) inherits the operator’s spectral accuracy while holding the late-time behaviour steadier. The PINO without data (Figure 4.10) still holds the low band, but its undershoot sets in earlier in the spectrum than in either operator trained with data.

4.5 Temporal Stability from the Physics Residual

The temporal growth of the FNO error in Figure 4.8 has a specific and instructive cause, and isolating it shows where the physics term makes a visible difference. As Section 2.4.2 explains, the operator applies the Fourier transform along the temporal axis as well as the spatial one, implicitly assuming the trajectory is periodic in time. It is not. The solution at $t = T$ does not match at $t = 0$, introducing jumps, so the assumption manufactures an artificial discontinuity at the temporal boundary and excites the Gibbs

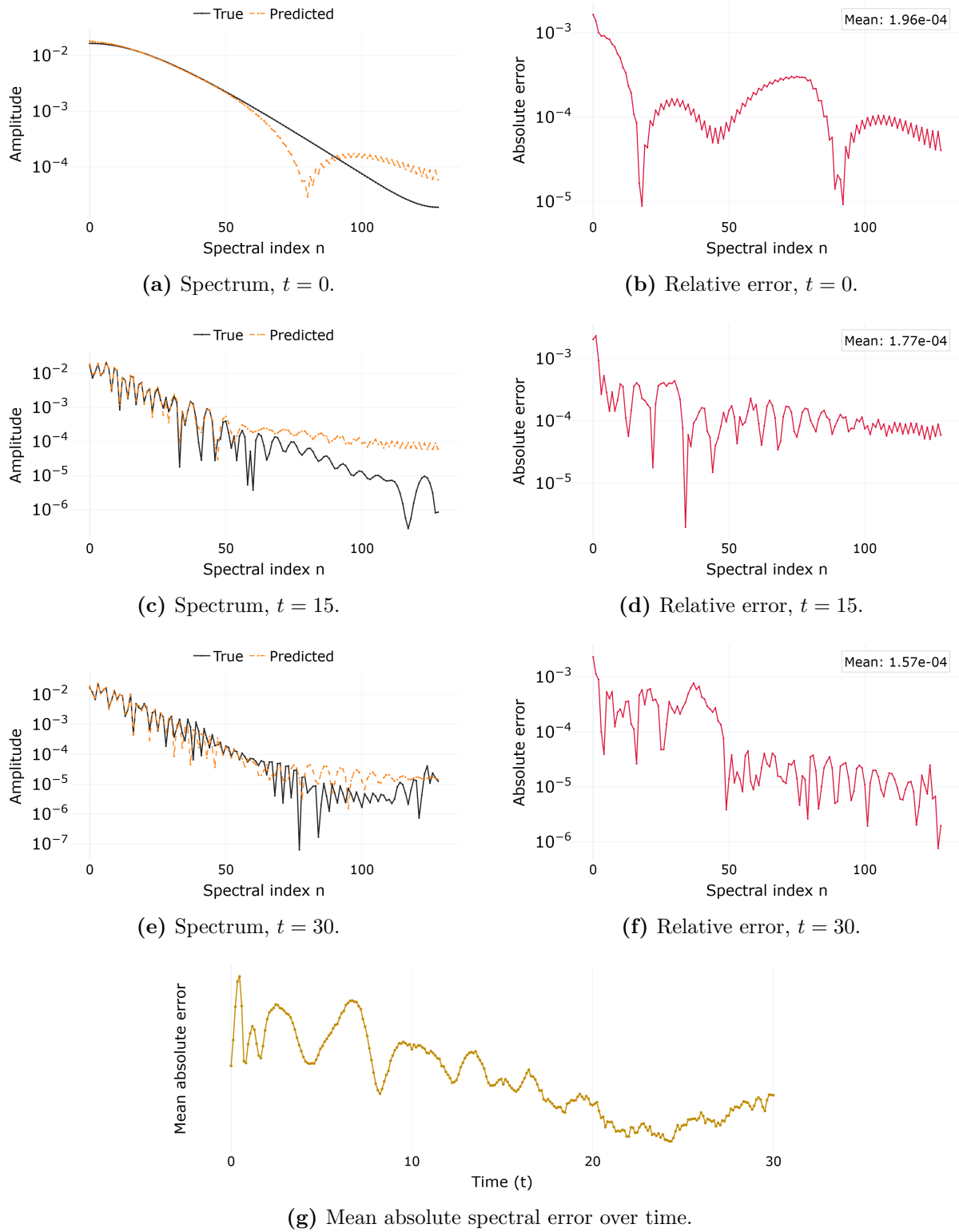


Figure 4.5 – MLP (pure data) spatial spectrum at $\alpha = \beta = 3.21$. Fitting the reference with no equation and no physics term, the network reproduces the low-wavenumber crests but undershoots the tail. The error stops falling where the reference keeps falling, so the two meet in the upper half of the spectrum with a clear case of spectral bias, with nothing else in the objective to obscure it.

Source: The author (2026).

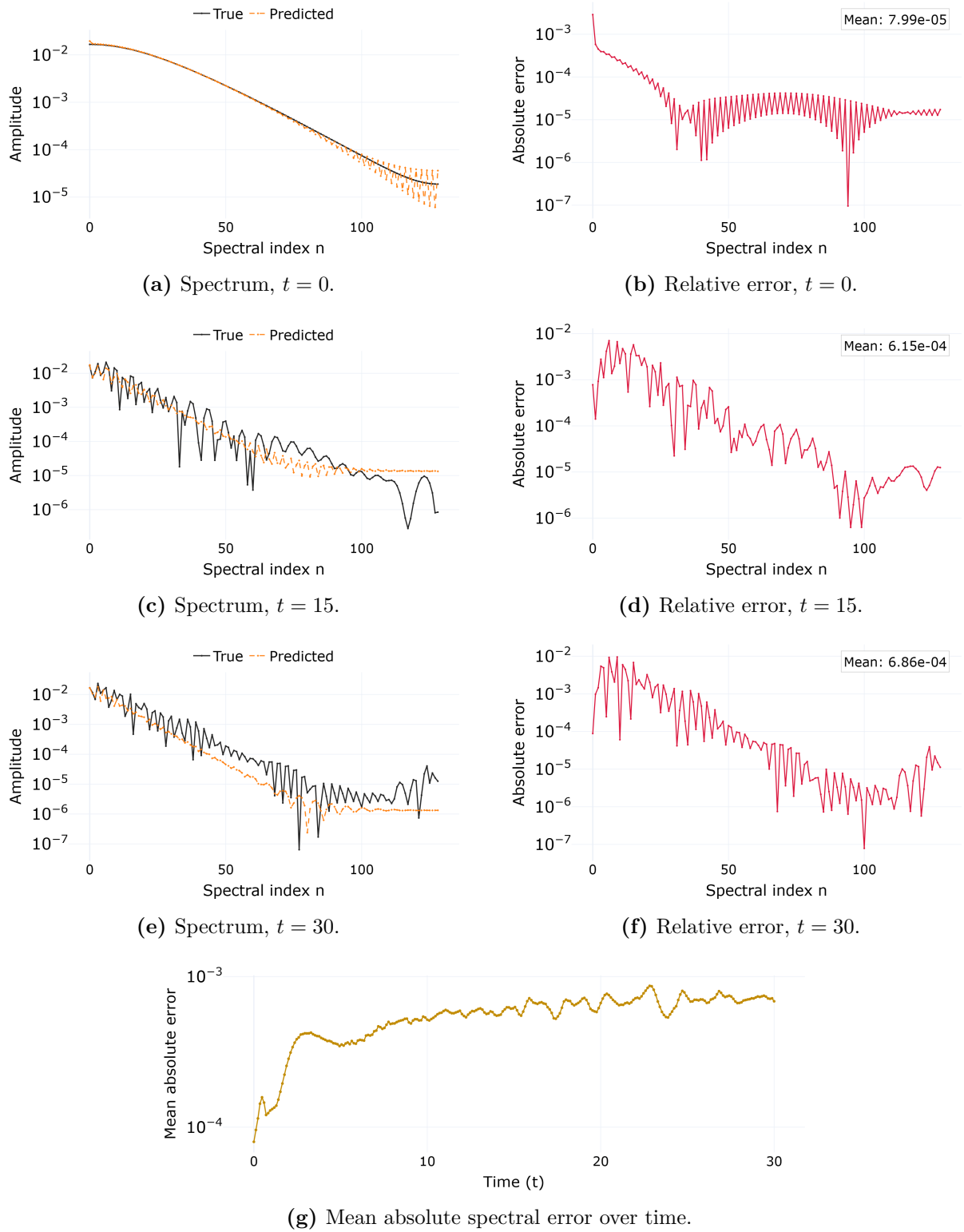


Figure 4.6 – PINN (pure physics) spatial spectrum at $\alpha = \beta = 3.21$. The network cannot represent the sech^2 tail even at $t = 0$, and past the first few modes the error follows the envelope of the reference itself, so the prediction contributes almost nothing at those wavenumbers. The agreement worsens as time advances.

Source: The author (2026).

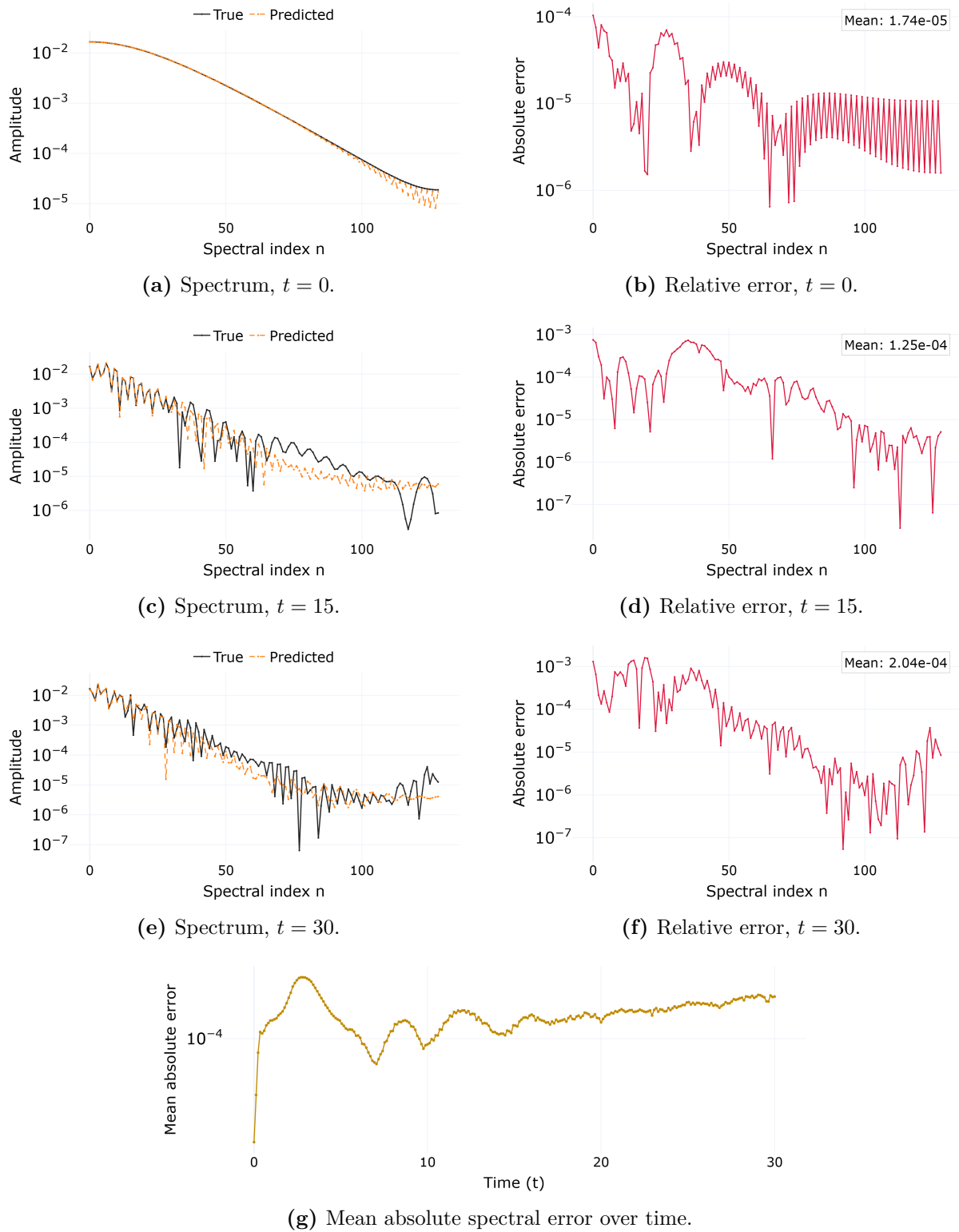


Figure 4.7 – PINN (data and physics) spatial spectrum at $\alpha = \beta = 3.21$. The low-wavenumber crests are tracked more tightly than in Figure 4.6, but the high-wavenumber undershoot remains and the error still grows in time.

Source: The author (2026).

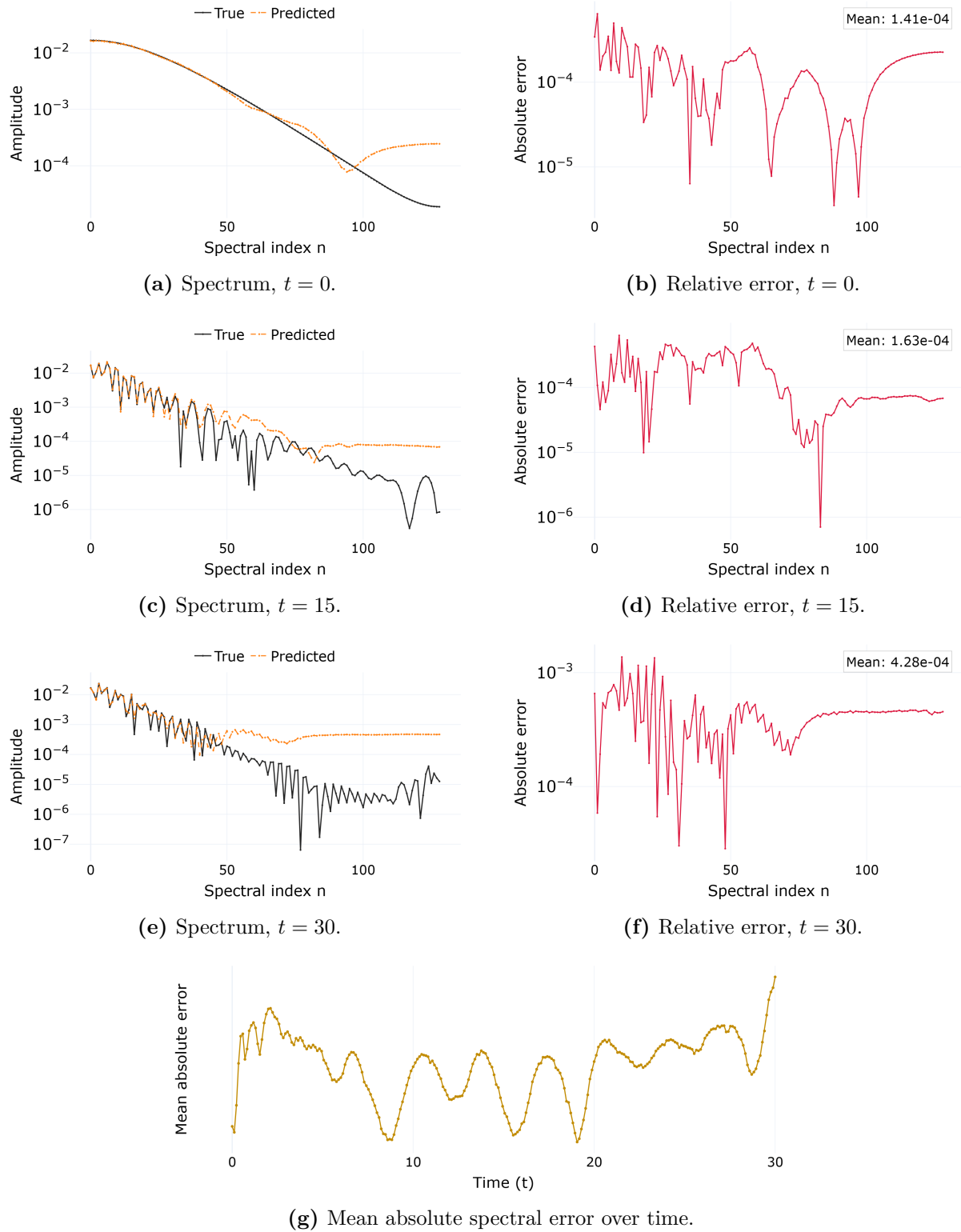


Figure 4.8 – FNO (pure data) spatial spectrum at $\alpha = \beta = 3.21$. Low and mid wavenumbers are tracked closely, but the error migrates to the high modes and grows in time.

Source: The author (2026).

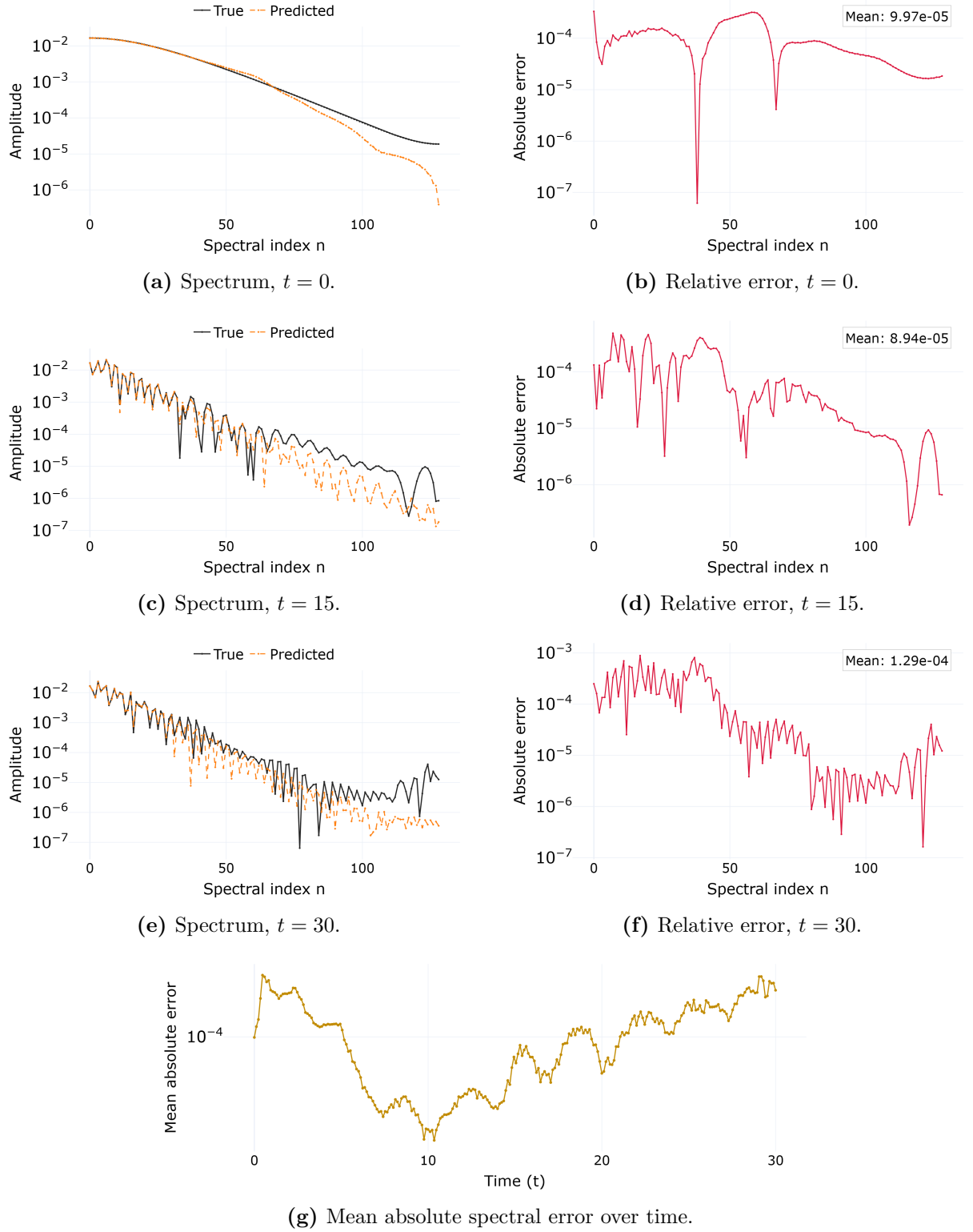


Figure 4.9 – PINO (data and physics) spatial spectrum at $\alpha = \beta = 3.21$. The prediction tracks the reference across the band, and the physics residual steadies the temporal evolution, so the fit does not degrade toward late time as sharply as the FNO's.

Source: The author (2026).

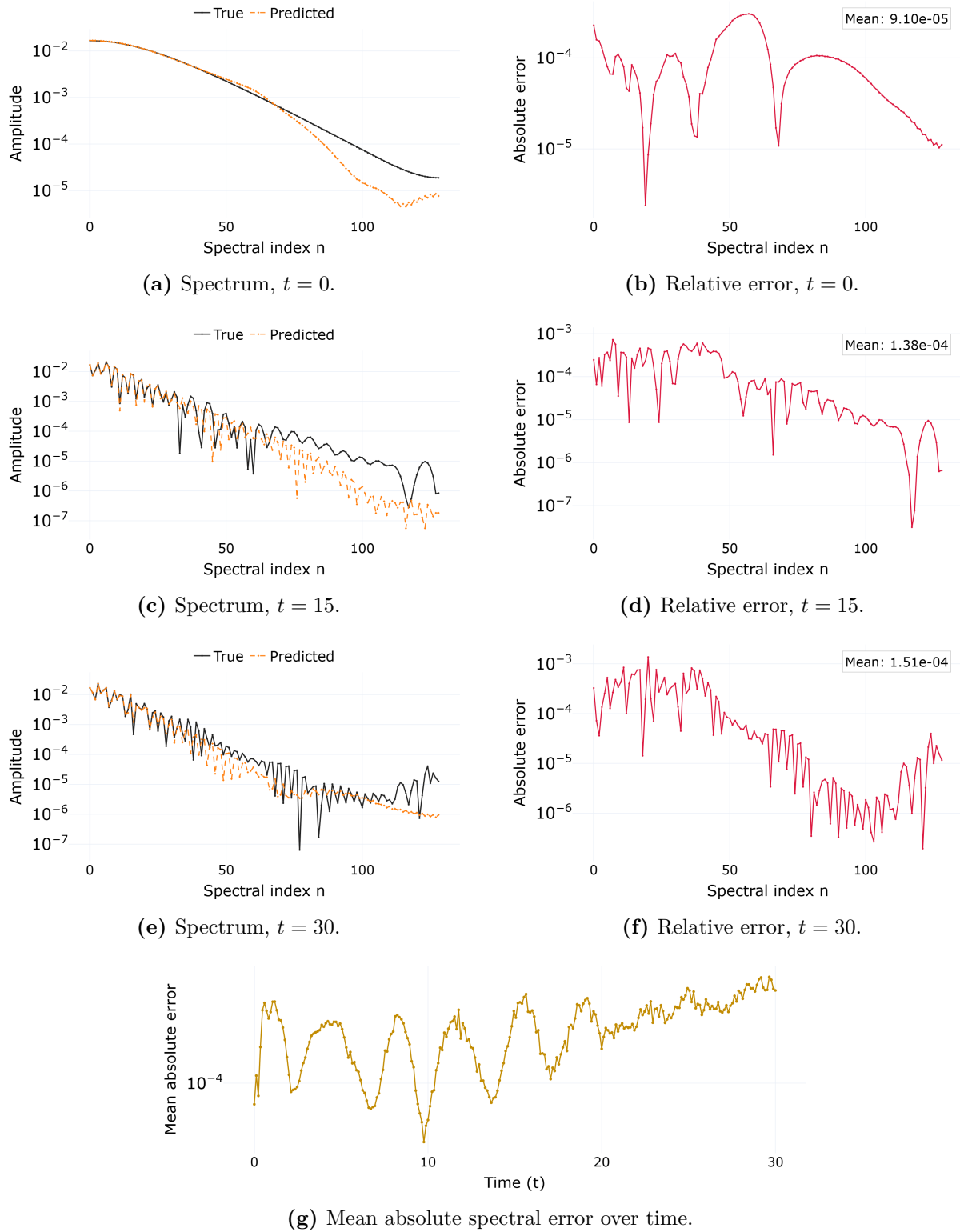


Figure 4.10 – PINO (pure physics) spatial spectrum at $\alpha = \beta = 3.21$. Without the supervised term the undershoot sets in earlier in the spectrum, confirming that the physics residual alone does not overcome spectral bias.

Source: The author (2026).

phenomenon there (GIBBS, 1899). The error curves of Figures 4.1 and 4.2, drawn earlier for the parameter sweep, make the effect and its correction visible once they are read along the time axis rather than across parameters.

In the FNO panel (Figure 4.1) a spike appears at all three nonlinearities and is the dominant part of the FNO’s late-time error. In the PINO panel (Figure 4.2) the predicted surfaces are visually identical, yet no such feature appears at the final time. The PDE residual and the initial-condition term penalize the temporal inconsistency that a periodic transform would otherwise tolerate. Suppressing the FNO’s Gibbs artifact is a concrete benefit of the physics term in this study, although the correction is a suppression and not a cure and the ideal implementation would not take temporal transforms.

4.6 Numerical Evidence for Spectral Bias

Derived in Section 2.3.4, this effect predicts that gradient training resolves low frequencies before high ones at a rate set by the eigenvalues of the Neural Tangent Kernel. Our experiment of Section 3.5 puts that prediction on the Boussinesq system in its barest form with a network from $\mathbb{R}$ to $\mathbb{R}$ learning the single profile $\eta(x, 15)$ at $\alpha = \beta = 3.21$, at three widths, by just gradient descent at the per-width step $\mu = 0.4/\lambda_{\max}$, the largest initial NTK eigenvalue, run for half a million iterations. Figure 4.11 reports the four checks.

Each panel tests one prediction. In Figure 4.11a the amplitude of the error along each eigendirection follows the predicted $|1 - \mu\lambda_k|^n$ of (2.38) nicely.

The measured curves leave the prediction once a coefficient reaches the noise floor and turns back up, where the drifting Jacobian overtakes the initial-kernel approximation. Late in the run they lift further as that drift accumulates. Figure 4.11b reads the same dynamics in Fourier space, where the lowest sampled wavenumber is driven down while the higher ones lie flat across the run. Figure 4.11c takes the whole spectrum instead of four sampled wavenumbers, averaging the absolute error over the low, mid and high bands defined in (3.10).

Figure 4.11d shows that the discrete crossing count of (2.47) carries no width, so if it really works the three networks must fall on one diagonal regardless of their size. For the reachable modes they do, across several iterations. The highest modes cross earlier than the initial-kernel count predicts, falling slightly below the diagonal, and they do so by more for the wider networks. That gap is a clear sign that the kernel does not stay frozen, and the diagnostic below traces it to its cause.

Figure 4.12 sets what the three runs produced against the profile they were asked to reproduce, and follows the widest network across the run.

Everything above assumes the kernel stays put. Figure 4.13 reports whether it

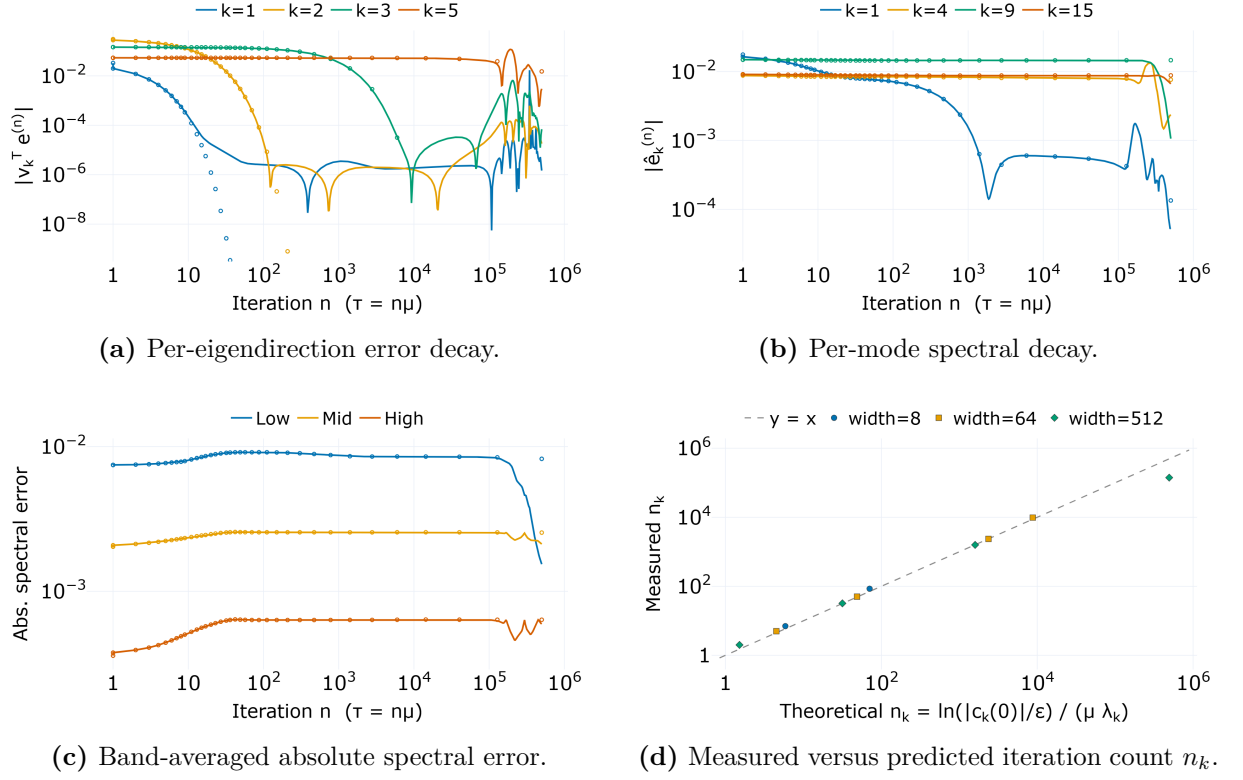


Figure 4.11 – Spectral-bias predictions measured in the minimal experiment of Section 3.5, an $\mathbb{R} \rightarrow \mathbb{R}$ network fitting $\eta(x, 15)$ at $\alpha = \beta = 3.21$ by full-batch gradient descent at the per-width step $\mu = 0.4/\lambda_{\max}$, run for 500,000 iterations. Panels (a)–(c) are the widest network, $N_{\text{MLP}} = 512$; panel (d) collects all three widths. Lines are measured, open markers are the initial-kernel prediction.

Source: The author (2026).

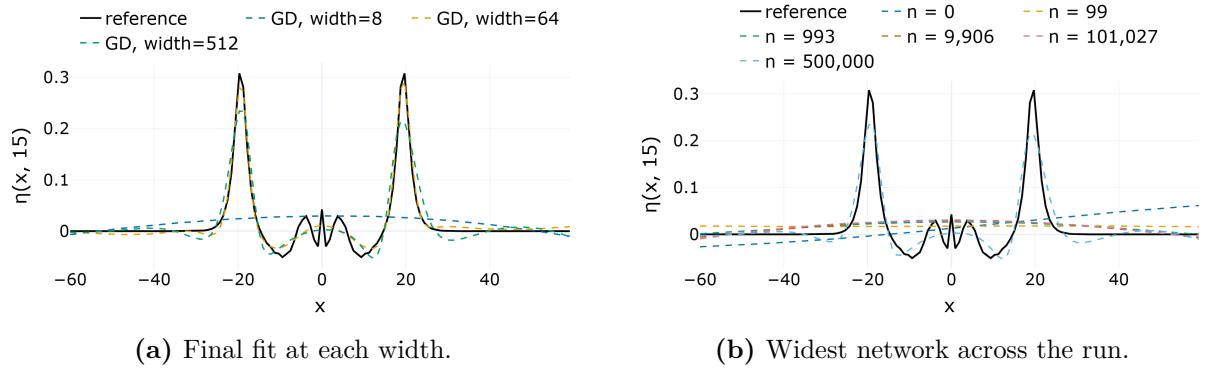


Figure 4.12 – The minimal networks against the reference profile $\eta(x, 15)$ at $\alpha = \beta = 3.21$. Panel (a) shows the final fit at each width; panel (b) follows the widest network across logarithmically spaced checkpoints. The narrowest network never leaves a nearly flat curve, while the wider ones recover the crests, and only at the very end of a half-million-iteration run.

Source: The author (2026).

actually does, measured along the same three runs.

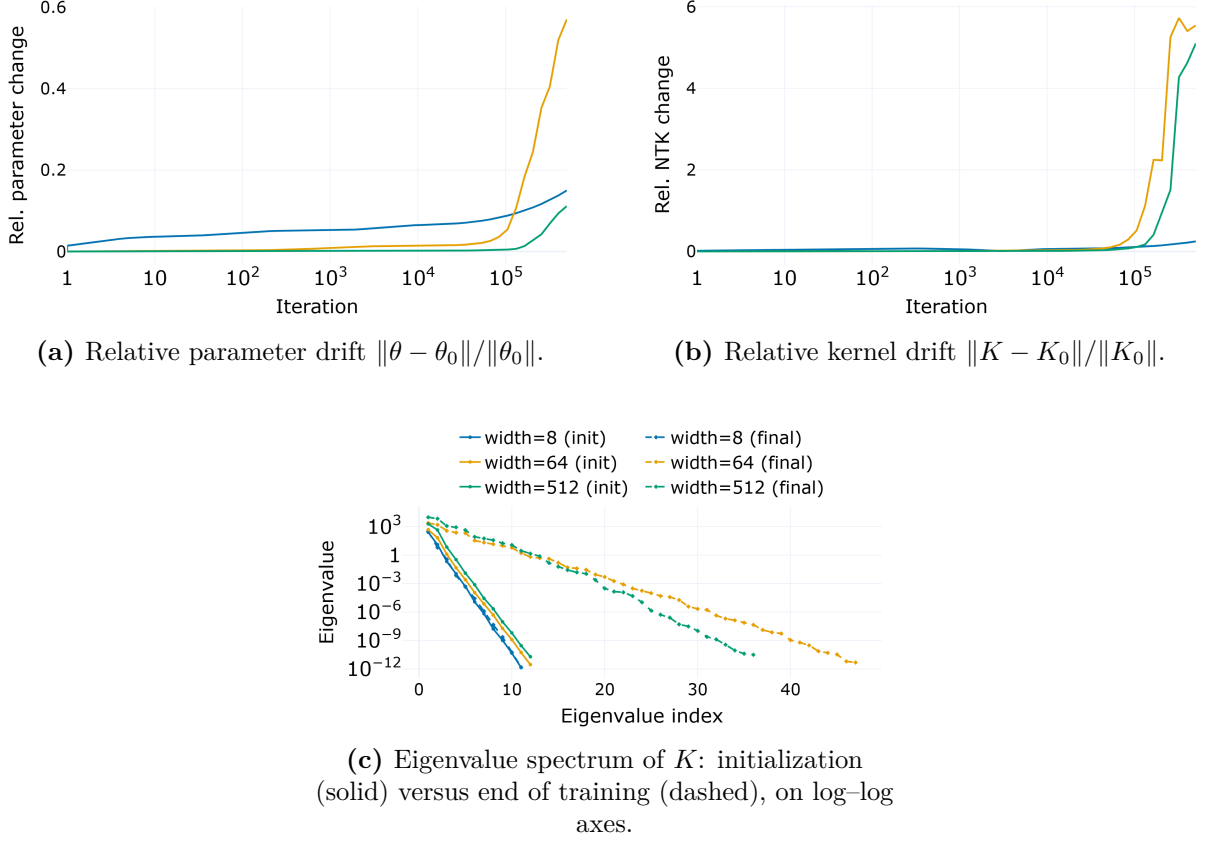


Figure 4.13 – Neural Tangent Kernel diagnostic for the minimal experiment at $\alpha = \beta = 3.21$, logged along the same gradient-descent runs as Figure 4.11, for widths 8, 64 and 512. Eigenvalues below the double-precision floor are numerical noise and are not drawn.

Source: The author (2026).

The parameters and the kernel do not settle in order of width. For the narrowest network both move only slightly and then hold (Figures 4.13a, 4.13b), while the two wider ones are quiet only at first and end far from where they began.

This is the eigenvalue disparity that Wang, Yu, et al. (2022) identify as the source of spectral bias. The adaptive and spectrum-aware optimizers of Section 5.3 are aimed at mitigating it.

The narrowest network’s spectrum barely changes over training and it stays in that gap; the wider ones flatten theirs by the end of the run, lifting the high modes clear of the floor and reaching frequencies a fixed kernel could not reach, showing the need for drift.

One expectation follows for the architectures of the next section: they inherit the same steep spectrum and so begin biased toward low frequencies.

4.7 Learning by Frequency Band

We now watch spectral bias develop as the optimizer runs. Figure 4.14 tracks the absolute spectral error in the low, mid, and high bands of (3.10) across training for all six methods, every one monitored at the common parameter $\alpha = \beta = 3.21$. The panels are therefore directly comparable, and each shows how one architecture and one regime meet the bias.

The panels share one feature: the high band is the hardest to move. Only the FNO brings it down by more than a modest fraction; elsewhere it ends within a factor of two of the magnitude of the reference, and in one case above it. In the PINN without data, the low band never falls below the reference either, so the worst of its error stays at low wavenumbers. The NTK spectrum of Figure 4.13 and the minimal experiment of Figure 4.11 predict this, now measured inside the full architectures.

The MLP panel (Figure 4.14a) shows that, since no physics term and no initial-condition term compete against the data fit, the low band is driven down while the high band barely moves, rising through most of the run before falling back to a little under where it started. This is the prediction of Section 2.3.4 but seen for a higher-dimensional MLP.

Holding the architecture fixed and adding the supervised loss gives the same picture in both the PINN and the PINO families. Data pulls the low and mid bands down markedly, while the high band barely shifts. The pure-physics panels leave the low and mid bands above what the same architecture reaches with data, and the high band close to where an untrained network would leave it. Data moves the resolvable bands; the physics constraint does little.

The FNO is the one place where the wall is partly breached, reaching the lowest low- and mid-band errors in the figure and a high band several times lower than any other method's, yet still far from its own low band. Read across the six panels, Figure 4.14 makes the general point that the operators reduce spectral bias rather than remove it, even the plain FNO leaves a high-frequency residual, and every other method leaves more.

This picture matches the one anticipated in Chapter 1 that these methods complement the pseudospectral solver that sets the accuracy ceiling here rather than replacing it. Their value is in amortizing the parameter sweep and in the temporal regularization physics supplies, and it is smallest at the high wavenumbers, where the reference solver is unchallenged. Overall, this shows that physics-informed interpolation can be a somewhat reliable way of approximating the continuous solution, but fitting physics alone must be taken with a grain of salt.

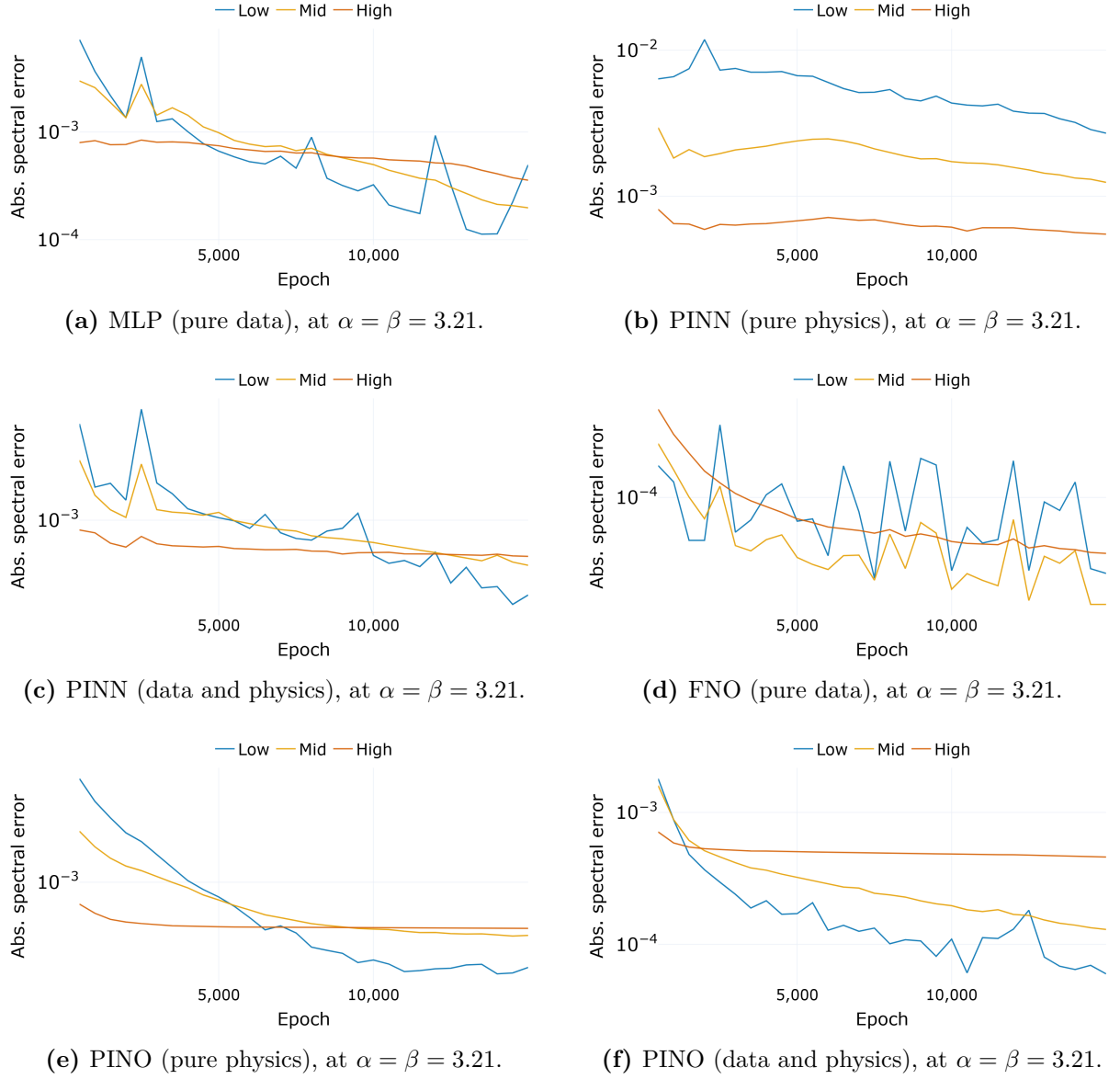


Figure 4.14 – Relative spectral error by frequency band during training, every method tracked at the common parameter $\alpha = \beta = 3.21$. The low band (blue), mid band (yellow) and high band (red) follow (3.10). In every method but the PINN without data the three bands end in the order the theory predicts, the low band lowest and the high band highest. That PINN is the exception to where the worst error sits: it is the only method whose low band never falls below the magnitude of the reference, so its largest error stays at low wavenumbers rather than high. That gap between the bands is the signature of spectral bias.

Source: The author (2026).

5 CONCLUSION

After an extensive set of tests, we can finally draw some general and specific conclusions. First, spectral bias is undeniably present in all architectures, as a general behavior of training dynamics. The data supervision helps approximate the low and mid spectral bands better than what physics-informed showed. On the other hand, physics-informed brought more stability on the time axis since it helps to mitigate the Gibbs artifact coming from Fourier transforms on non-periodic time. The sections below read that result family by family, and then set out the boundaries of the study and the work that should follow it.

5.1 Assessment of Each Method Family

The four architectures split into two families by how one trained network relates to the parameter axis, and reading them family by family also answers the questions posed in Chapter 1.

Neural Networks: MLP and PINN

Recall that this family of methods aims to learn the solution for only one parameter. The MLP (a “pure data PINN”) shows spectral bias in its plain form, fitting spectra as training goes. The pure physics PINN contrasts with it in the sense that it does not fit well mid and high bands of the spectrum: PINNs do not seem to approximate wave profile tails, meaning the high frequency content. Thus, the combination data + physics seems to be the best, gathering both strengths, the physics-informed part of the loss working as a regularization term a priori of the solution.

Neural Operators: FNO and PINO

In this family, we have one architecture capable of showing how the solution looks at any model parameter close to what it saw in the dataset. Here we have similar results. The FNO, the pure data supervised architecture, simply learns to fit and, while still showing spectral bias, has its main weakness in the Gibbs phenomenon anomaly from

taking Fourier transforms in the time axis. This introduces error shootouts at the temporal boundaries. Meanwhile, the pure physics PINO presents the same multiparameter solution, but shows the same behavior as the pure physics PINN, meaning they cannot approximate high frequency details of the solution or, in other words, their ripples. The best method is, again, the mix of data + physics, and here we can observe that the Gibbs anomaly in the FNO is suppressed in this supervised PINO, ultimately presenting this work’s best architecture, which fits the data and respects the model’s physics. Also, discretization invariance holds for the two architectures, with only one caveat: increasing resolution yields slightly lower quality results, but the architectures remain advantageous as surrogates.

Spectral Bias Study

After we gathered, to the best of our knowledge, theoretical findings that explain or expose spectral bias, we condensed them in this work and designed the experiments to verify its plausibility. For all we can say, while the simple network struggled to learn the target, the kernel stayed close to its initial setting and thus our theory predicts the loss behavior well on each NTK kernel eigenvector direction (or mode, for our problem). The network size also shows us that, really, increasing the size of the MLP indeed helps the kernel to change less, allowing for the good estimates presented, such as loss evolution by mode and iteration a priori estimate. As long as the kernel does not drift (drifting is a good thing, it means the network is learning), our predictions are good.

The cost accounting of Table 4.1 completes the comparison. A pointwise network is the cheaper object to train and the more expensive one to use, since a new parameter means a new training run, while an operator pays its training cost once and answers the rest of the sweep in forward passes. Both physics-informed variants cost more to train than their data-driven counterparts, since every iteration differentiates the network to form the residual.

5.2 What This Study Cannot Claim

Several boundaries fence in the results above, and they should be weighed before the conclusions are carried anywhere else.

Every experiment used one initial condition, the unit solitary profile of Table 3.1, and held the two coefficients equal. The operators therefore learned a one-parameter family and not the four-parameter Boussinesq family classified by Bona et al. (2002) and Martins (2023), so the generalization reported here is generalization along a single coordinate. Decoupling α from β , or varying the initial data, is a different potential

challenge to be explored.

The temporal advantage measured for the PINO is entangled with an implementation choice. Our operators transform along the time axis as well as the spatial one, and the terminal-time artifact of Section 4.5 follows from that assumption of temporal periodicity. An operator that marched in time, as Li et al. (2021) suggest, would not produce the artifact at all, so part of what the residual is credited with here is the repair of a defect the architecture introduced.

Capacity and budget were fixed rather than tuned. Both operators keep 16 Fourier modes per axis, every trained method ran modest 15,000 Adam iterations at a single learning rate, and the whole pipeline ran from one seed on one GPU.

The reference solution is itself a numerical solution, accurate to the order of the pseudospectral scheme on the grid it was solved on and not exact, which matters at the high wavenumbers, where it carries the largest share of its own error and where the trained methods are weakest.

The minimal experiment is a model of the mechanism and not a model of the trained methods. It fixes time, fits a single profile by plain gradient descent, and works with a kernel small enough to be eigendecomposed at every checkpoint. Not much of this analysis transfers easily to PINNs and NOs.

5.3 Considerations and Directions Forward

For directions: adaptive weighting of the loss terms (WANG; TENG, et al., 2021), the Neural Tangent Kernel weighting derived for PINNs by Wang, Yu, et al. (2022), and the second-order and spectrum-aware optimizers of Khodakarami et al. (2026) are possible near-future steps to consider, evaluating possible mitigations for spectral bias.

An operator that marches in time removes the temporal periodicity assumption by construction, which would isolate how much of the late-time steadiness of the PINO comes from the physics term itself. A related question is whether the residual keeps helping once the training data are made scarce, the regime the physics-informed formulation was proposed for and the one this study, with a reference solution available at every point of the grid, did not test. Maybe we could try some real problems with observations that include errors and whatnot.

The decoupling of the nonlinearity from the dispersion, varying the initial data, and extending to the two-dimensional system would show whether the ordering found here survives a solution operator of higher dimension, and the analysis of Krishnapriyan et al. (2021) gives reason to expect the physics-informed losses to grow harder to optimize as that happens.

We can also explore other architectures like DeepONet (LU et al., 2021) and hybrid

schemes that hand the high-frequency residual back to a classical corrector, which would test whether the complementary role identified here can be built into a single solver.

At this time, we can accept spectral bias as a behavior we can choose to mitigate, and we recommend doing so, since the literature presents simple and effective ways. It should be taken into account and acknowledged, if the reader is aiming to implement these architectures. This also shows that ML methods do not ultimately come to take the place of numerical methods. Key Fluid Dynamics simulations in the industry will not stop using classical-based solvers for ML methods, as these should actually work together. In the end, a trained operator answers a parameter test sweep for parametric physical models in a single forward pass, for a fraction of what repeated solves would cost. Also, unlike classical methods, theoretical analyses for ML methods are much harder to come by and not yet fully consolidated.

All of this considered, we finish this work noting that this is a novel field and most certainly the new era of Fluid Dynamics where ML models promise fast evaluation and new predictions. Of course this is not tied to only neural networks and neural operators, given the vastness of the field.

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